

Neurocritical Care for Patients With Ischemic Stroke

By T. M. Leslie-Mazwi, MD

REVIEW ARTICLE



CONTINUUM AUDIO
INTERVIEW AVAILABLE
ONLINE

ABSTRACT

OBJECTIVE: Management of stroke due to large vessel occlusion (LVO) has undergone unprecedented change in the past decade. Effective treatment with thrombectomy has galvanized the field and led to advancements in all aspects of care. This article provides a comprehensive examination of neurologic intensive care unit (ICU) management of patients with stroke due to LVO. The role of the neurocritical care team in stroke systems of care and the importance of prompt diagnosis, initiation of treatment, and continued monitoring of patients with stroke due to LVO is highlighted.

LATEST DEVELOPMENTS: The management of complications commonly associated with stroke due to LVO, including malignant cerebral edema and respiratory failure, are addressed, stressing the importance of early identification and aggressive treatment in mitigating negative effects on patients' prognoses. In the realm of medical management, this article discusses various medical therapies, including antithrombotic therapy, blood pressure management, and glucose control, outlining evidence-based strategies for optimizing patient outcomes. It further emphasizes the importance of a multidisciplinary approach to provide a comprehensive care model. Lastly, the critical aspect of family communication and prognostication in the neurologic ICU is addressed.

ESSENTIAL POINTS: This article emphasizes the multidimensional aspects of neurocritical care in treating patients with stroke due to LVO.

INTRODUCTION

The emergent management of large vessel occlusion (LVO) stroke¹ has changed immensely in the past decade. The introduction of highly effective thrombectomy techniques, proven effective in multiple, international randomized trials in 2015, transformed the field of stroke.² Since those positive trials, we have witnessed the expansion of stroke candidacy through randomized trials that address two important expansion groups: (1) patients presenting after 6 hours from onset of symptoms (late window)^{3,4} and (2) patients with large pretreatment infarct cores (large core).⁵ Trials are actively assessing other subgroups, including patients with distal vascular occlusion and patients with LVO but limited clinical deficits. There has been a rapid evolution of thrombectomy techniques and available

CITE AS:

CONTINUUM (MINNEAP MINN)
2024;30(3, NEUROCRITICAL CARE):
611-640.

Address correspondence to
Dr T. M. Leslie-Mazwi, 1959 NE
Pacific St, RR 650 HSB, Seattle
WA 98195, tml01@uw.edu.

RELATIONSHIP DISCLOSURE:

Dr Leslie-Mazwi has received personal compensation in the range of \$500 to \$4999 for serving on a scientific advisory or data safety monitoring board for IQVIA Inc and in the range of \$5000 to \$9999 for serving on scientific advisory or data safety monitoring boards for Koninklijke Phillips N.V., WorldCare Clinical, and Zoll Medical Corporation.

UNLABELED USE OF PRODUCTS/INVESTIGATIONAL USE DISCLOSURE:

Dr Leslie-Mazwi reports no disclosure.

© 2024 American Academy
of Neurology.

devices. We have seen the advent of novel thrombolytics because tenecteplase is increasingly becoming the preferred recombinant tissue plasminogen activator (rtPA) over alteplase in stroke.⁶ In concert with these developments, intense attention has been directed toward patient triage and systems of care. As with all aspects of stroke care, thrombectomy is subject to disparities in access and utilization. These disparities exist with regard to race, ethnicity, geography, sex, and gender.^{7–11} There are vast differences globally in access to thrombectomy, postthrombectomy care, and rehabilitation.¹² Uneven utilization of best practices impacts patient outcomes and represents a clear and crucial target for future improvements to stroke systems of care.

This article summarizes the role that the neurologic intensive care unit (ICU) ideally plays in the management of patients with LVO. For a detailed discussion of neuroendovascular therapies and the management of other types of strokes, refer to the April 2023 issue of *Continuum*, “Cerebrovascular Disease.” For an overview of the myriad stroke syndromes, refer to the article “The Neurocritical Care Examination and Workup” by Sarah Wahlster, MD, and Nicholas J. Johnson, MD,¹³ in this issue of *Continuum*.

THE ROLE OF THE NEUROCRITICAL CARE UNIT IN STROKE SYSTEMS OF CARE

An effective stroke center requires both neurocritical care staff and a dedicated neurocritical care or advanced stroke unit space to optimize patient care and subsequent outcomes. Neurointensivists should be closely involved in discussions about stroke care systems with institutional and regional leadership. The active participation of the neurocritical care team in the treatment of patients who have had a stroke is required for the certification of the centers as comprehensive stroke centers or thrombectomy-capable stroke centers.^{14,15}

The neurocritical care team typically cares for patients with LVO after thrombectomy and those ineligible for thrombectomy, playing an important role in triage and acceptance for LVO transfers. Given the time-sensitivity of stroke due to LVO, rapid and coordinated patient transfer is imperative. Emergency department overcrowding and other barriers can slow the transfer process, making the neurologic ICU a preferred destination for LVO transfers.¹⁶ However, in some hub-and-spoke networks, patients with LVO may be accepted directly to the angiography suite to improve arrival-to-puncture times and potentially 90-day outcomes.¹⁷ These transfers still generally require the involvement of the neurocritical care team, particularly if patients have other critical care needs aside from their stroke.

Understanding factors that drive transfer outcomes and inefficiencies is important to improve thrombectomy access; the neurologic ICU can be instrumental in overcoming these challenges. Data for transfer times indicate that a delay of 2 hours or more from presentation to transfer at local hospitals is common,¹⁸ which allows adequate time to prepare for patient arrival, subsequent rapid assessment, and performance of necessary imaging and initiation of relevant therapies for the stroke itself or its complications. Even patients with stroke due to LVO who are not candidates for thrombectomy can benefit from neurologic ICU care. Transfers that are excluded from thrombectomy consideration are usually excluded because of infarct growth leading to large established infarct (49%) or for mild symptoms despite the occlusion (33%), both of which often require neurologic ICU admission.¹⁹

INTERDISCIPLINARY COMMUNICATION IN ACUTE STROKE CARE

Patients with LVO are seen by several teams before they are evaluated by the neurocritical care team, creating multiple points of patient handoff.

Admission to the Neurologic Intensive Care Unit

For patients with LVO who undergo thrombectomy, there are handoffs between the emergency department, acute stroke team, anesthesia, and neuroendovascular services, and then information is finally given to the neurocritical care team. For patients who are not candidates for thrombectomy, handoffs occur between the emergency department, stroke, and neurocritical care teams. Structured handoffs ensure that necessary information is conveyed and patient safety is assured. It is beneficial to establish a consistent structure for communication regarding all patients with LVO.²⁰

In addition to ensuring smooth communication between teams, it is important to facilitate an accelerated transfer of patients with LVO to the neurologic ICU from both the emergency department and the angiography suite. Although emergency department length of stay is not clearly an indicator of stroke care in the emergency department because of the variability of patient treatment routes,²¹ it is noteworthy that among critically ill patients who have had a stroke, an emergency department length of stay of 5 hours or longer before neurologic ICU transfer has been independently associated with poor outcomes at hospital discharge.²²

Transfer out of the Neurologic Intensive Care Unit

When patients with LVO no longer need to be in the neurologic ICU, there must be a thorough handoff to the stroke or inpatient team receiving the patient. As with all handoffs, structured and complete communication ensures patient safety during the transition. A neurologic ICU transfer checklist (TABLE 3-1²³) can decrease hospital length of stay and improve clinicians' perception of patient safety. The checklist should include (1) a brief summary of patients' medical histories and ICU experiences, (2) a list of pending tasks, and (3) an organ system–focused inventory of inpatient medical issues.²³

MONITORING PATIENTS WITH STROKE DUE TO LARGE VESSEL OCCLUSION

After admission to the neurologic ICU, patients with stroke due to LVO require close monitoring of the neurologic examination, hemodynamics (particularly blood pressure), glucose, and temperature. All patients should be monitored using standard hemodynamic monitoring, including cardiac telemetry. Routine use of other monitors is usually unnecessary, although may be considered; there is growing interest in a range of real-time monitoring techniques, particularly for cerebral perfusion.²⁴ These tools may have future application in the setting of limited neurologic examinations in patients who are intubated and sedated.

Neurologic Examination

As described in detail in the article “The Neurocritical Care Examination and Workup” by Sarah Wahlster, MD, and Nicholas J. Johnson, MD,¹³ in this issue of *Continuum*, it is imperative to closely monitor the neurologic examination for all neurocritical care patients, including those with stroke due to LVO. The use of pupillometry can facilitate standardized pupillary assessment with increased

KEY POINTS

- The active participation of the neurocritical care unit in the care of patients with stroke is required for certification as comprehensive stroke centers or thrombectomy-capable stroke centers.
- Structured handoffs between the emergency department and the neurocritical care team ensure that necessary information is conveyed and patient safety is assured for patients with stroke due to large vessel occlusion (LVO).
- After admission to the neurologic intensive care unit (ICU), patients with stroke due to LVO require close monitoring of the neurologic examination, hemodynamics (particularly blood pressure), glucose, and temperature.

reliability, which is particularly helpful for comatose patients. The purpose of serial examinations is to detect clinical changes that indicate a new neurologic event. The neurologic examination is usually performed every 15 minutes immediately after thrombectomy, then it is spaced to every 1 to 2 hours by 8 hours after thrombectomy. Deescalation of examination frequency is supported by data demonstrating that examination changes (and hemorrhage) typically occur relatively early.²⁵

Blood Pressure

After stroke due to LVO, the fate of the ischemic penumbra largely depends on maintaining perfusion above the threshold for infarction. Therefore, optimizing blood pressure remains a potential target to improve neurologic outcomes in patients with LVO during the acute and subacute stages. There is a U-shaped relationship between blood pressure and outcome in patients who have had a stroke; both extremes of blood pressure have prognostic significance for death and disability.²⁶ Although patients with LVO frequently present with elevated blood pressure, which improves after recanalization, elevated blood pressure both before and after thrombectomy is associated with adverse outcomes for patients with stroke due to LVO.^{27,28}

There is uncertainty about the ideal blood pressure target in patients after they have had a thrombectomy or in patients who are not recanalized. Further,

TABLE 3-1

Components to Include in a Comprehensive Neurologic Intensive Care Unit to Floor Transfer Checklist^a

Component	Purpose
Synthesis of intensive care unit course	Brief summary of the patient's medical history and intensive care unit experience; includes key dates; is usually best maintained prospectively for admitted patients and may take the form of a transfer note
Pending action items	Tests, imaging, consults, and social or disposition-related items
Systems-based review Examination fluctuations, hemodynamic stability, cardiac arrhythmias, secretions and aspiration risk, oxygen and respiratory support, volume status, Foley catheter and urinary retention, antibiotics (indication, duration, end date), deep vein thrombosis (DVT) prophylaxis, dysphagia and oral administration status, bowel regimen, glucose management, home medication reconciliation, code status, expected disposition on discharge	Detailed list, including "if-then" plan articulated for each relevant point; not all items will apply to each patient
Family communication and goals of care	Summary of what has been discussed with family and how they are coping

^a Data from Murray NM, et al, Neurohospitalist.²³

it is unclear whether systolic, diastolic, or mean arterial pressure is the most important value. In 2021, evidence from randomized trials demonstrated that intensive systolic blood pressure reduction to between 100 mm Hg and 129 mm Hg after successful thrombectomy did not reduce hemorrhagic conversion rates at 36 hours after procedures compared with a standard-care systolic blood pressure target of 130 mm Hg to 185 mm Hg, although between-group differences in actual achieved blood pressure was small.²⁹ Beyond absolute values, dynamic fluctuations in blood pressure have been identified as a strong prognostic marker after stroke due to LVO and noted to be associated with an increased risk of intracranial hemorrhage following IV thrombolytics.³⁰

With this context, systolic blood pressure should be kept at less than 185 mm Hg, and excessive variability or hypotension should be avoided after successful recanalization or thrombolytic exposure. For patients without successful recanalization or who are not candidates for thrombectomy, permissive hypertension preserves the ischemic penumbra,³¹ and allowing autoregulation to a systolic blood pressure of 220 mm Hg or supporting hemodynamics with vasopressor therapy to prevent hypotension is reasonable.

Glucose

After a stroke, both hypoglycemia and hyperglycemia are problematic. Hypoglycemia impairs oxidative metabolism, leading to brain dysfunction and compromise of already injured tissue.³² Hyperglycemia exacerbates intracellular acidosis and increases cerebral edema, infarct size, and infection rates. Hyperglycemia in the initial 24 hours after stroke correlates with adverse outcomes, including increased risks of reperfusion injury and hemorrhagic transformation.³³

Although monitoring blood glucose in patients with stroke due to LVO is important, optimum targets are uncertain. The SHINE (Stroke Hyperglycemia Insulin Network Effort) trial demonstrated no significant improvement in the 90-day outcome from intensive glucose control (between 80 mg/dL and 130 mg/dL) using insulin infusion compared with standard control (less than 180 mg/dL) with intermittent subcutaneous insulin. Moreover, the insulin infusion increased the risk of hypoglycemia and the frequency of monitoring by nursing.³⁴ Current American Heart Association/American Stroke Association (AHA/ASA) guidelines suggest meticulous blood glucose monitoring, treatment of hyperglycemia to maintain glucose levels between 140 mg/dL and 180 mg/dL, and prompt correction of hypoglycemia (less than 60 mg/dL).³⁵ IV short-acting insulin is preferred over oral agents in the acute setting after stroke because of dosing flexibility, duration of action, and lower risk of hypoglycemia.

Temperature

Hyperthermia after stroke due to LVO is common, with temperatures greater than 37.5°C (99.5°F) reported in up to one-third of patients.³⁶ Higher body temperature is associated with worse outcome after stroke, given the impact of fever on the ischemic cascade and neuronal injury. An infective cause (most typically aspiration pneumonitis or pneumonia) should always be sought and treated. Antipyretic medications and nonpharmacologic methods of fever control (eg, use of icepacks, surface cooling pads, or endovascular cooling) should be implemented to maintain normothermia. While data from animal

KEY POINTS

- The purpose of serial neurologic examinations after thrombectomy is to detect clinical changes that indicate a new neurologic event. The neurologic examination is usually performed every 15 minutes immediately after thrombectomy, then it is spaced to every 1 to 2 hours by 8 hours after thrombectomy.
- There is a U-shaped relationship between blood pressure and outcome in patients who have had an ischemic stroke; both extremes of blood pressure have prognostic significance for death and disability.
- Current American Heart Association/American Stroke Association guidelines suggest meticulous blood glucose monitoring, treatment of hyperglycemia to maintain glucose levels between 140 mg/dL and 180 mg/dL, and prompt correction of hypoglycemia (less than 60 mg/dL) in patients who have had an ischemic stroke.

models support the neuroprotective effect of induced hypothermia, there is currently no clinical benefit (and potentially harm) in humans, and this intervention should not be used routinely for patients with LVO in the neurologic ICU.³⁷ The role of hypothermia, however, remains an area of active investigation.

POSTTHROMBECTOMY COMPLICATIONS FOR PATIENTS WITH STROKE DUE TO LARGE VESSEL OCCLUSION

Multiple randomized trials have demonstrated the safety of thrombectomy, with complication rates ranging from 0.6% to 6% in appropriately selected patients. **TABLE 3-2**³⁸⁻⁴¹ details some of the major complications after thrombectomy, with management essentials that are important for the neurocritical care team.

After thrombectomy, the percutaneous endovascular access site must be monitored closely, particularly in patients who are hypocoagulable after thrombolytic administration.⁴² Patients who have had stroke often have femoral,

TABLE 3-2 Select Complications After Thrombectomy in Patients With Stroke Due to Large Vessel Occlusion^a

Location	Complication	Management essentials
Access site	Groin hematoma and bleeding from femoral access site	Manual pressure, ultrasonography to evaluate for pseudoaneurysm
	Femoral artery pseudoaneurysm	Ultrasound-guided compression, thrombin injection, possible open surgical repair
	Femoral artery dissection	Usually observation, rarely stenting is necessary
	Femoral artery occlusion	Evaluation by vascular surgery or interventional radiology for revascularization (endovascular or open surgical)
	Retroperitoneal hematoma	Manual pressure on arteriotomy, emergent blood cross-matching and usually transfusion, CT and CT angiography of abdomen and pelvis, possible endovascular or surgical repair
	Distal hand ischemia (for radial access)	Vascular surgery consultation
Extracranial	Cervical artery dissection	Usually observation, antiplatelet therapy, rarely carotid or vertebral stenting or sacrifice if vessel compromise is severe
	Vasospasm	Usually observation, often intraarterial vasolytic medication (verapamil, nicardipine, milrinone, nimodipine)
Intracranial	Vessel perforation	Management in the neuroendovascular suite (including possible vessel sacrifice), postprocedure CT imaging, intensive care unit management of hemorrhagic complication
	Embolism to new territory	Management in the neuroendovascular suite (including possible additional thrombectomy or intraarterial thrombolytic), intraprocedure and postprocedure hemodynamic manipulation to assist perfusion
	Distal embolism	Management in the neuroendovascular suite (including possible additional thrombectomy or intraarterial thrombolytic), intraprocedure and postprocedure hemodynamic manipulation to assist perfusion.

^a Data from Leslie-Mazwi T, et al, J Neurointerv Surg³⁸; Krishnan R, et al, Neurology³⁹; Pilgram-Pastor SM, et al, Neurointerv Surg⁴⁰; and Hapipi Ngankou E, et al, Stroke.⁴¹

as opposed to radial, access sites, because of the large bore of base catheters and thrombectomy devices, but procedural outcomes are similar between the two access locations.⁴³ Radial access is safer and should be prioritized if it does not compromise the thrombectomy.⁴⁴ In rare cases, direct carotid access may be used, but this carries a risk of cervical hematoma.⁴⁵ Postprocedural monitoring of the access site differs between radial and femoral locations, but the focus in both cases is on complication avoidance, detection, and management.⁴⁶

Radial Access

After angiography is performed via radial access, patients typically have a compressive band applied to their wrists, which is gradually deflated after the procedure by the bedside neurologic ICU nurse. In the event of bleeding from the radial site, the band is reinflated until bleeding ceases and then kept inflated for a longer period. There usually is no dressing at the radial access site, although a bandage may be applied. These patients have no restrictions on positioning or use of the hand related to the access site. Distal hand ischemia is extremely uncommon because of the collateral blood supply to the hand via the superficial and deep palmar arches.⁴⁷

Femoral Access

After angiography via femoral access, it is usual practice to apply a closure device to the arterial access site to seal the arteriotomy.^{48,49} Manual pressure is applied in the angiography suite to assist with hemostasis before the site is dressed. Despite this, patients with a femoral access site are at risk for bleeding. Postprocedure orders require the neurocritical care team to perform regular groin checks and evaluations to ensure distal limb perfusion is intact (ie, the temperatures of the limb and distal pulse sites are unchanged), initially at a frequency of every 10 to 15 minutes and then gradually spaced out. Patients must keep the hip joint relatively straight for 4 to 6 hours after procedures to decrease the risk of groin-site hematoma.³⁸ This may delay the extubation of patients who remain intubated after thrombectomy because they are unable to sit up. For patients at high risk of aspiration when supine, the entire bed can be elevated 30 degrees to 40 degrees in the reverse Trendelenburg position.

Patients with a femoral access site are at risk for limb ischemia, groin site hematoma, or retroperitoneal hematoma. Perfusion concerns should prompt emergent vascular surgery consultation. In the event of hemorrhage at the femoral access site, immediate manual pressure must be applied. Hypotension after thrombectomy via femoral access should raise concern for retroperitoneal hemorrhage, and in addition to acute resuscitation, immediate manual pressure should be applied to the groin, and the neuroendovascular team should be contacted. These patients need blood matched for possible transfusion and typically require abdominal and pelvic imaging to evaluate for active bleeding that could necessitate assistance from interventional radiology or vascular surgery.⁵⁰

HEMORRHAGIC TRANSFORMATION AFTER STROKE DUE TO LARGE VESSEL OCCLUSION

Serial imaging is performed for patients with stroke due to LVO to evaluate for treatment complications (most specifically hemorrhagic transformation) after thrombolytics or thrombectomy and to determine final infarct volume. The perceived risk of hemorrhage determines the timing

KEY POINTS

- Higher body temperature is associated with worse outcomes after stroke given the impact of fever on the ischemic cascade and neuronal injury.
- After thrombectomy, postprocedural monitoring of the percutaneous endovascular access site differs between radial and femoral locations, but the focus in both cases is on complication avoidance, detection, and management.

and frequency of follow-up imaging. Patients who have had thrombolytic treatment or thrombectomy should get imaged if there is neurologic deterioration or concern for complication (eg, vessel perforation with contrast extravasation), but routine imaging should also be repeated within 24 hours.

Reperfusion injury and hemorrhage can be viewed as a continuum, determined by the pathologic permeability of the blood-brain barrier. Infarct size is a reliable predictor of hemorrhagic transformation.⁵¹ Ischemia leads to increased permeability of the blood-brain barrier through activation of a variety of enzymes,⁵² the most prominent being the matrix metalloproteinases, in particular, matrix metalloproteinase-9.⁵³ This process is exacerbated by circulating thrombolytic drugs.⁵⁴ Reperfusion of ischemic cerebral arteries contributes to the risk of hemorrhage, with the highest risk at hypertensive extremes (blood pressure greater than 220/105 mm Hg).

The primary significance of hemorrhage is its impact on recovery. To define this effect, there are several classification systems for hemorrhagic

TABLE 3-3

Heidelberg Classification of Hemorrhagic Transformation After Stroke^a

Class	Type	Clinical significance
1: Hemorrhagic transformation of infarcted brain tissue		
1a	Hemorrhagic infarction 1: scattered small petechiae, no mass effect	Minimal; no impact on outcomes
1b	Hemorrhagic infarction 2: confluent petechiae, no mass effect	Minimal; no impact on outcomes
1c	Parenchymal hematoma 1: hematoma within infarcted tissue occupying <30% of infarct bed, minimal mass effect	Moderate; worsened short-term recovery, but unchanged 90-day outcomes
2: Intracerebral hemorrhage within and beyond infarcted brain tissue		
	Parenchymal hematoma 2: hematoma occupying 30% or more of the infarcted tissue, with mass effect	Significant; worse 90-day outcomes
3: Intracerebral hemorrhage outside the infarcted brain tissue		
3a	Parenchymal hematoma remote from infarct	Variable impact on patient recovery depending on severity of hemorrhage
3b	Intraventricular hemorrhage	Variable impact on patient recovery depending on severity of hemorrhage
3c	Subarachnoid hemorrhage	Variable impact on patient recovery depending on severity of hemorrhage
3d	Subdural hemorrhage	Variable impact on patient recovery depending on severity of hemorrhage

^a Modified with permission from von Kummer R, et al, Stroke.⁵⁵ © 2015 Wolters Kluwer Health.

transformation, with varying definitions of symptomatic intracranial hemorrhage. Refinements and expansion have occurred with the Heidelberg Bleeding Classification (**TABLE 3-3**), developed in 2015⁵⁵ and validated against older scales.⁵⁶ This classification incorporates the previously widely used ECASS I (European Cooperative Acute Stroke Study I) classification⁵⁷ in addition to expanding it to include other forms of poststroke and postintervention hemorrhage.

After angiography, more than 80% of patients with LVO will have hyperdensities due to contrast on postprocedure CT imaging, which are distinct from hemorrhage. A higher admission National Institutes of Health (NIH) Stroke Scale (NIHSS) score, thrombolytic exposure, hyperglycemia, and a large contrast load are prominent risk factors for hyperdensities due to contrast, which is more likely to be seen in women than men.^{58,59} Strategies to distinguish between contrast extravasation and hematoma include serial noncontrast CT imaging, MRI with hemorrhage-sensitive sequences (eg, gradient recalled echo [GRE]), and dual-energy CT. Dual-energy CT, which reliably differentiates between tissues with similar x-ray attenuation but different compositions with regard to atomic numbers, is particularly useful within 24 hours after thrombectomy. Imaging using this modality can change the diagnosis from hematoma to contrast staining in up to 34% of patients and alter the hemorrhage grade in up to 10%.⁶⁰

Although distinct from hemorrhage, the presence of contrast staining is not benign because it represents both potentially infarcted tissue and tissue at risk for hemorrhagic transformation. Both contrast staining and hemorrhage are associated with worse outcomes (odds ratio of 11.3 and 10.4, respectively), and contrast staining is associated with delayed hemorrhagic transformation (odds ratio of 4.5).⁶¹ Areas of contrast staining that represent infarction have higher densities on iodine mapping, representing a higher degree of blood-brain barrier disruption, which results in contrast enhancement. Cutoff values of 34.1 Hounsfield units were predictive of this risk.⁶² Iodine concentration greater than 1.35 mg/dL has been associated with the later development of hemorrhage, with an area under the receiver operating characteristic curve of 0.89.⁶³ Dual-energy CT can therefore provide both diagnostic and prognostic information. **CASE 3-1** describes a patient with hemorrhagic transformation after stroke due to LVO.

Although ideal blood pressure management remains a topic of intense debate for patients who have had a stroke, it seems likely that persistently elevated arterial pressures in the context of hemorrhagic transformation may be detrimental, leading to continued hemorrhage, repeat hemorrhage, and edema. Maintaining a systolic blood pressure of less than 140 mm Hg or 160 mm Hg is reasonable following the detection of or suspicion for hemorrhage; the use of rapidly acting IV antihypertensive agents helps to balance the risk for hemorrhage against maintenance of adequate cerebral perfusion. Identification of contrast staining should also inform the postprocedural approach to blood pressure management, although blood pressure targets are unclear in this patient population.

Reversal of Thrombolytic Effect

Thrombolytic administration can lead to bleeding (intracranial or systemic) in 6% to 10% of patients; bleeding can occur beyond the duration of drug action because of the persistence of a hypocoagulable state.⁶⁴ Bleeding may manifest as

KEY POINTS

- Serial imaging is performed for patients with stroke due to LVO to evaluate for treatment complications (most specifically hemorrhagic transformation) after thrombolytics or thrombectomy and to determine the final infarct volume.
- Strategies to distinguish between contrast extravasation and hematoma in patients who have had stroke due to LVO include serial noncontrast CT imaging, MRI with hemorrhage-sensitive sequences (eg, gradient recalled echo [GRE]), and dual-energy CT.
- There are no widely accepted approaches to the reversal of thrombolytic-related hemorrhage after thrombolytic administration (for either alteplase or tenecteplase), and institutional protocols vary.

CASE 3-1

A 66-year-old man presented 3 hours after symptom onset of right hemiparesis with a National Institutes of Health (NIH) Stroke Scale score of 19. CT showed a hyperdense left middle cerebral artery (FIGURE 3-1A). CT angiography confirmed left middle cerebral artery occlusion (FIGURE 3-1B). CT perfusion imaging (FIGURE 3-1C) showed core infarct and penumbral tissue, with a mismatch ratio of 2:1, indicating that penumbral volume was double the volume of the established infarct. The patient underwent a thrombectomy. Initial angiographic runs in the arterial and parenchymal phase demonstrated middle cerebral artery occlusion (FIGURE 3-1D) and a large parenchymal defect. After successful clot retrieval, final angiographic runs demonstrated reperfusion, with a full parenchymal phase on final angiography (FIGURE 3-1E).

Head CT immediately after thrombectomy (FIGURE 3-2A) demonstrated hyperdensity in the region of the previous parenchymal defect on angiography. Dual-energy head CT (FIGURE 3-2B) showed hyperdensity that demonstrated suppression with application of the iodine suppression sequence consistent with contrast extravasation, without frank hematoma. Head CT at 24 hours after thrombectomy (FIGURE 3-2C) showed new hyperdensity consistent with tissue hematoma in the areas of previous contrast staining and a sizable left middle cerebral artery infarct with mass effect (parenchymal hematoma 2 in the Heidelberg hemorrhagic conversion classification). Hemicraniectomy was not required, and cerebral edema was managed medically. As a result of the hemorrhagic conversion of the stroke, the patient had pneumatic compression devices but was not given pharmacologic venous thromboembolism prophylaxis. On day 5 of admission, the patient became acutely hypoxic. CT of his chest with a pulmonary embolism protocol (FIGURE 3-3A) revealed multiple pulmonary emboli. The patient underwent placement of an inferior vena cava filter for protection until anticoagulation could be commenced on day 10 after thrombectomy (FIGURE 3-3B). There was no worsening of the hemorrhage. Subsequently, the patient was transferred out of the neurologic intensive care unit for ongoing inpatient care.

COMMENT

This case highlights that, even when reperfusion is achieved early, patients with limited collateral circulation can experience large infarcts. Initial head CT changes proved predictive of hemorrhagic conversion of the infarct. Parenchymal hematoma 2 criteria mean the patient will likely have a worsened 90-day outcome because of the hemorrhage. Pharmacologic deep venous thrombosis (DVT) prophylaxis was appropriately not given, despite the patient being at high risk of DVT formation, and after the development of a pulmonary embolism, therapeutic anticoagulation was too risky because of the danger of worsening cerebral hemorrhage. A delay in initiating systemic anticoagulation that balanced these competing risks to the patient was determined, with the inferior vena cava filter offering protection from life-threatening embolism in the interval.

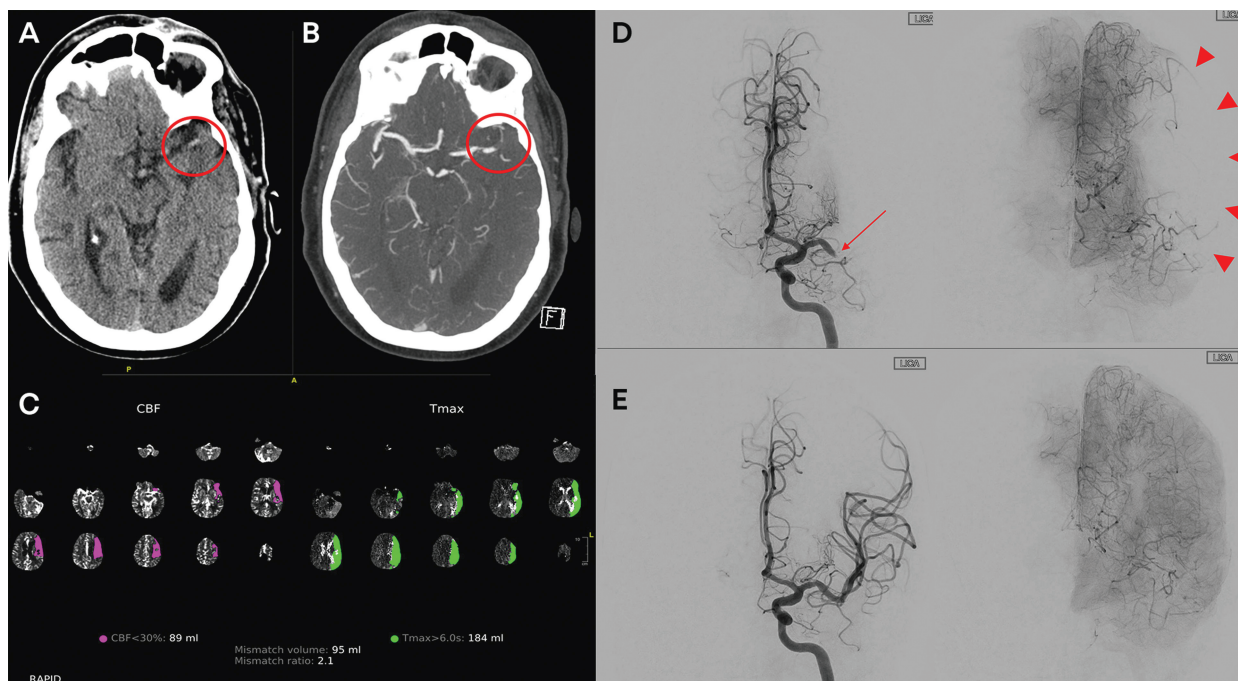
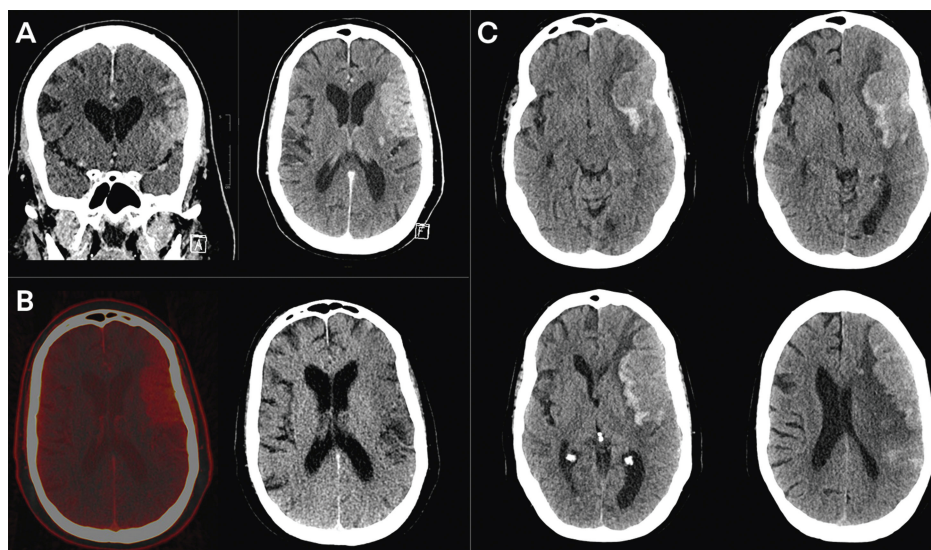


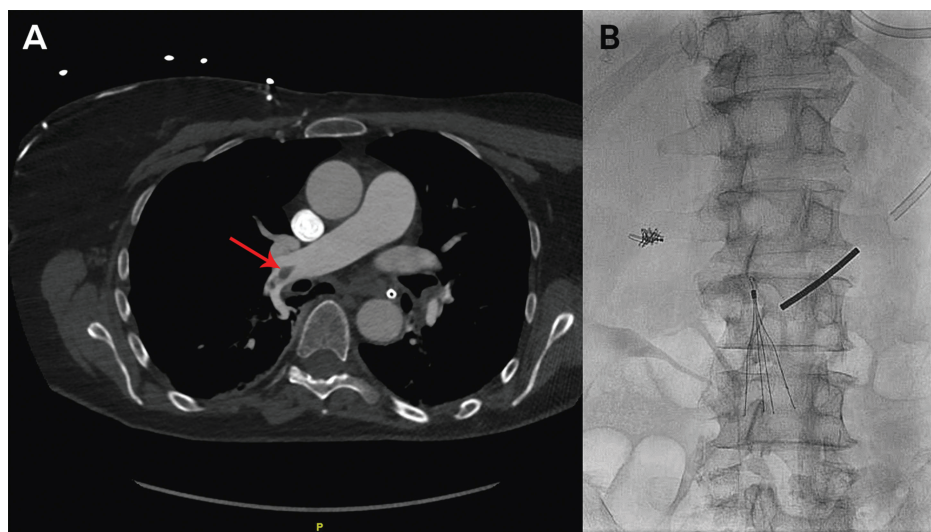
FIGURE 3-1

Hemorrhagic transformation after stroke due to large vessel occlusion in the patient in **CASE 3-1**. **A**, CT shows a hyperdense left middle cerebral artery (red circle). **B**, CT angiography confirms left middle cerebral artery occlusion (red circle). **C**, CT perfusion imaging shows core infarct (purple shading) and penumbral tissue (green shading) with a mismatch ratio of 2:1, indicating that penumbral volume was double the volume of the established infarct. **D**, After thrombectomy, digital subtraction angiography, anteroposterior view, with initial angiographic runs in the arterial and parenchymal phase shows middle cerebral artery occlusion (arrow) and a large parenchymal defect (arrowheads). **E**, After successful clot retrieval, digital subtraction angiography, anteroposterior view, with final angiographic runs shows reperfusion, with a full parenchymal phase on final angiography.

CONTINUED ON PAGE 622

CASE 3-1CONTINUED FROM
PAGE 621**FIGURE 3-2**

Hemorrhagic transformation after stroke due to large vessel occlusion in the patient in **CASE 3-1**. **A**, Coronal and axial head CT immediately after thrombectomy shows hyperdensity in the region of the previous parenchymal defect on angiography. **B**, Axial dual-energy head CT shows hyperdensity that demonstrates suppression with application of the iodine suppression sequence consistent with contrast extravasation without frank hematoma. **C**, Axial head CT at 24 hours after thrombectomy shows new hyperdensity consistent with tissue hematoma in the areas of previous contrast staining and a sizable left middle cerebral artery infarct with mass effect (parenchymal hematoma 2 hemorrhagic conversion).

**FIGURE 3-3**

Hemorrhagic transformation after stroke due to large vessel occlusion in the patient in **CASE 3-1**. **A**, Five days after admission, axial chest CT with a pulmonary embolism protocol reveals multiple pulmonary emboli (red arrow). **B**, The patient underwent placement of an inferior vena cava filter for protection until anticoagulation could be commenced on day 10 after thrombectomy.

a neurologic examination change if it is intracranial or as systemic hypotension if from an occult bleeding site such as the gastrointestinal tract, retroperitoneum, chest, or thigh, or, more dramatically, as hemoptysis or severe epistaxis. If a thrombolytic infusion (alteplase) is still running when bleeding is detected, it should be stopped. Bleeding within the first 24 hours after administration should prompt consideration of emergent thrombolytic reversal. However, there are no widely accepted approaches to the reversal of thrombolytic-related hemorrhage for either alteplase or tenecteplase, and institutional protocols vary. **FIGURE 3-4** summarizes the conceptual approach to postthrombolytic hemorrhage and treatment options that should form the basis of institutional protocols.

BRAIN EDEMA AFTER STROKE DUE TO LARGE VESSEL OCCLUSION

Infarcted tissue can lead to edema due to the failure of membrane transporters with the influx of sodium and water into the necrotic or ischemic cells. The risk of secondary injury makes this an important neurocritical care consideration. Although edema may affect recovery, even for patients with small infarcts,⁶⁵ clinically significant cerebral edema is only seen in large-territory cerebral or cerebellar infarcts. This can present in three different ways: (1) a rapid and fulminant course (within 24 to 36 hours), (2) a gradually progressive course (over several days), or (3) an initial worsening course followed by a plateau and resolution (about a week).⁶⁶ Swelling typically peaks on day 3, but more protracted courses can occur. Because edema after stroke due to LVO is

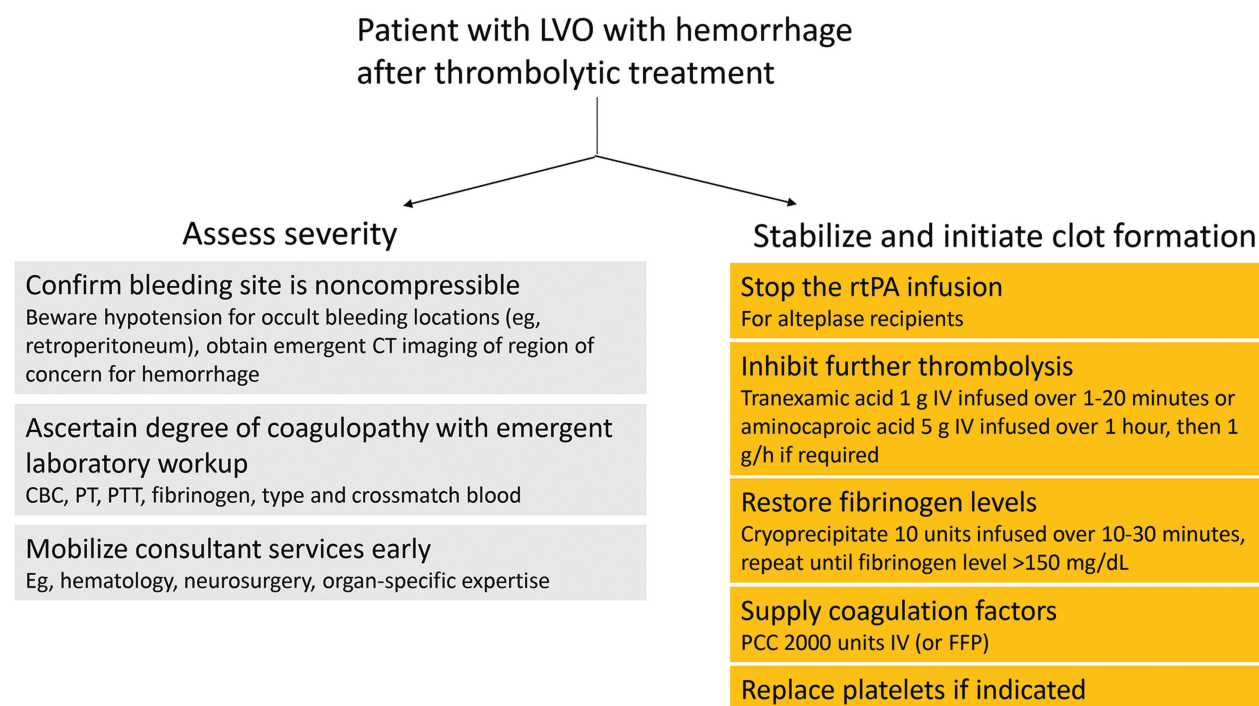


FIGURE 3-4

Conceptual approach to recombinant tissue plasminogen activator (rtPA)-related hemorrhage.

Note the parallel process of assessing severity and initiating reversal. The order of pharmacologic reversal is based on availability of agents and varies by center.

CBC = complete blood cell count; FFP = fresh frozen plasma; LVO = large vessel occlusion; PCC = four-factor prothrombin complex concentrate; PT = prothrombin time; PTT = partial thromboplastin time; rtPA = recombinant tissue plasminogen activator.

associated with high morbidity and mortality, it is referred to as *malignant cerebral edema*; it carries a mortality rate of almost 80% if untreated and more than 50% despite maximal medical management.

The development of malignant cerebral edema is most common in young patients, those with high NIHSS scores (20 or greater for dominant or 15 or greater for nondominant hemisphere lesions), onset of nausea or vomiting within 24 hours, systolic blood pressure greater than 180 mm Hg within 12 hours, and an early decrease in level of alertness.^{67,68} Findings on admission CT angiography that should raise concern about the development of malignant cerebral edema include the presence of a carotid T occlusion or incompleteness of the circle of Willis, exposing more territory to ischemic risk from a single occlusion.

When there is concern for the development of malignant cerebral edema, serial imaging should be obtained, beginning within the first 6 hours of admission. Early CT hypodensity, large final infarct volume (ie, greater than one-third of the middle cerebral artery territory), and early radiologic progression of midline shift (or posterior fossa crowding for cerebellar infarcts) are the key concerning radiologic findings.⁶⁹ Additional considerations are the location of the infarct, particularly the degree of temporal lobe involvement, given the risk of medialization and herniation of the uncus. MRI with diffusion-weighted imaging volume greater than 82 mL demonstrates high specificity (98%) but a sensitivity of only 52% for the development of malignant cerebral edema.⁷⁰ The addition of clinical examination parameters (specifically, NIHSS scores of 22 or greater) at 24 hours improves sensitivity to nearly 80%, while high specificity remains, allowing the clinical course to be accurately predicted in 93% of patients.⁷¹ Reperfusion is generally thought to not modify the risk of poststroke edema, although there may be benefit in patients with very large infarct volumes (greater than 130 mL).⁷² There is therefore a need to monitor for edema even after successful thrombectomy.

Despite concerns about secondary injury, intracranial pressure monitoring has no role in patients with stroke due to LVO who are at risk for malignant edema.⁶⁶ There are both medical and surgical options to manage poststroke cerebral and cerebellar edema.

Medical Management of Brain Edema

Despite several 2021 trials,⁷³ there have been limited developments in treatment options for cerebral edema. Patient positioning with the head of the bed raised above 30 degrees from supine and midline head positioning (if possible) is optimal. Given the potential contribution of hyponatremia to cerebral edema, the serum sodium goal for patients at risk for edema is eunatremia (135 mmol/L to 145 mmol/L). Once poststroke edema begins to develop, intermittent administration of hyperosmolar agents (ie, mannitol and hypertonic saline) is the mainstay of medical management. These agents have been studied in patients with traumatic brain injury⁷⁴ (refer to the article “Traumatic Brain Injury and Traumatic Spinal Cord Injury” by Jamie E. Podell, MD, and Nicholas A. Morris, MD, FAAN, FNCS,⁷⁵ in this issue of *Continuum*), but there are limited data for their use in stroke.⁷⁶

Mannitol is given as a 20% solution at a dose of 1 g/kg of body weight every 6 hours. Serum osmolality, serum urea nitrogen, sodium, and glucose should be monitored every 6 hours to allow for calculation of the osmolar gap. Recommendations vary for an osmolar gap that necessitates holding mannitol,

ranging from greater than 10 mOsm/kg to greater than 20 mOsm/kg, but an elevated gap suggests that mannitol has accumulated and additional doses should not be given until the gap is lower. Mannitol is renally cleared, so its use should be limited in those with acute or chronic kidney injury.

Hypertonic saline can be given as 30-mL boluses of 23.4% saline or 300-mL boluses of 3% saline every 6 hours. Serum sodium should be monitored every 6 hours because boluses are typically not given if sodium is greater than 160 mmol/L. Sustained hyponatremia via a continuous infusion of hypertonic saline may lessen the effect of bolus hyperosmolar therapy by reducing the gradient across which water can be pulled out of tissue, and it is generally avoided unless it develops as a consequence of bolus hyperosmolar therapy. Hypertonic saline expands intravascular volume, so it should be avoided in patients with congestive heart failure. Many hospital protocols require central access for hypertonic saline administration, so until this is obtained, mannitol is often used as the first-line agent.

In the event of abrupt clinical decompensation due to poststroke edema, intubated patients can be hyperventilated, targeting PaCO₂ between 30 mm Hg and 34 mm Hg, to lower intracranial pressure while preparing for surgical decompression. However, because of rapid (within hours) metabolic correction of respiratory alkalosis, it is well established that there is no benefit to sustained hyperventilation,⁷⁷ and eucapnia should be targeted for patients at risk for poststroke edema.

Surgical Management of Brain Edema

In the presence of intractable swelling after stroke, the removal of part of the skull may be necessary to give the swollen brain room to expand and relieve intracranial pressure. Several randomized trials confirm the effectiveness of hemicraniectomy in enhancing survival rates and outcomes after malignant edema following a hemispheric middle cerebral artery infarction in patients younger than 60 years.^{78–80} A more contemporary trial involving patients between the ages of 60 and 82 years indicated better survival rates, with the tradeoff of an increase in severe disability.⁸¹ Evidence therefore supports aggressive use of hemicraniectomy within 24 to 48 hours of the onset of stroke symptoms in patients aged 60 years or younger who have a large hemispheric infarction (ie, encompassing more than two-thirds of the middle cerebral artery territory) and a reduced level of consciousness. For patients between the ages of 60 and 80 years, hemicraniectomy remains a lifesaving procedure, but it should only be considered if patients and families are willing to accept a potentially high degree of disability. Technical considerations include the need for both hemicraniectomy and duraplasty to achieve true decompression, adequate size of the craniectomy defect (at least 12 cm × 9 cm), sparing of the midline (2 cm from vertex) to protect venous structures, and the extension of the defect to the floor of the middle fossa.⁸² For full or multiterritory infarcts, the additional measure of temporal lobectomy may be pursued.

There have been no randomized surgical trials for posterior fossa stroke and only limited prospective trials.^{83,84} However, given the serious repercussions of edema in this area, a suboccipital craniectomy is typically offered when there is concern for significant cerebellar infarction. The specific goal of surgery is relief of brainstem compression and secondary brainstem injury because recovery from cerebellar infarction is typically robust. Extrapolating from surgical

KEY POINTS

- Although edema may affect recovery, even for patients with small infarcts, clinically significant cerebral edema is only seen in large-territory cerebral or cerebellar infarcts. This can present in three different ways: (1) a rapid and fulminant course (within 24 to 36 hours), (2) a gradually progressive course (over several days), or (3) an initial worsening course followed by a plateau and resolution (about a week).
- In the event of abrupt clinical decompensation due to poststroke edema, intubated patients can be hyperventilated, targeting PaCO₂ between 30 mm Hg and 34 mm Hg, to lower intracranial pressure while preparing for surgical decompression.
- Evidence supports aggressive use of hemicraniectomy within 24 to 48 hours of the onset of stroke symptoms in patients aged 60 years or younger who have a large hemispheric infarction (ie, encompassing more than two-thirds of the middle cerebral artery territory) and a reduced level of consciousness.
- For patients between the ages of 60 and 80 years who have a large hemispheric infarction (ie, encompassing more than two-thirds of the middle cerebral artery territory) and a reduced level of consciousness, hemicraniectomy remains a lifesaving procedure, but it should only be considered if patients and families are willing to accept a potentially high degree of disability.

indications for cerebellar hemorrhage,⁸⁵ decompressive suboccipital craniectomy is offered, following cerebellar ischemia, to patients who show new cranial nerve symptoms, reduced consciousness, signs of brainstem compression, hydrocephalus, or lesions larger than 3 cm in diameter. These patients can clinically decompensate rapidly, so they must be monitored very closely with a low threshold for surgical intervention. **CASE 3-2** demonstrates the management of cerebellar edema after stroke due to LVO.

Patients who have hemicraniectomy require a helmet when they are out of bed and ultimately need to have their flap replaced in a delayed fashion, usually

CASE 3-2

An 82-year-old man on apixaban, with a history of atrial fibrillation, presented to his local hospital with onset of acute vertigo and headache as well as ataxia on examination (National Institutes of Health [NIH] Stroke Scale score of 1). Head CT (**FIGURE 3-5A**) demonstrated subtle evidence of a left cerebellar hypodensity consistent with infarct. CT angiography of the head and neck showed no large vessel occlusion, intracranial stenosis, or atherosclerosis. The patient continued to receive apixaban 5 mg daily and was given additional aspirin, 325 mg per local stroke protocol. Subsequent brain MRI (**FIGURE 3-5B** and **FIGURE 3-5C**) demonstrated a large left cerebellar infarct with suspected hemorrhagic conversion and mass effect on the fourth ventricle and scattered bilateral occipital infarcts consistent with an embolic etiology. The hemorrhage was classified as a parenchymal hematoma 1. The patient was transferred urgently to a comprehensive stroke center. Head CT performed 24 hours after initial presentation (**FIGURE 3-5D**) to evaluate increasing drowsiness demonstrated new effacement of the fourth ventricle, with new onset obstructive hydrocephalus. The patient was intubated, was given IV mannitol, and underwent urgent surgical decompression (**FIGURE 3-5E**). He was initially started on aspirin on day 7 while in the neurologic intensive care unit, and then he was transferred out of the intensive care unit with a transition plan to restart apixaban at 3 weeks from presentation.

COMMENT

Delayed diagnosis of posterior circulation stroke is common because of highly variable symptoms. CT imaging of the posterior fossa is further confounded by bony artifacts, making the detection of regions of ischemia difficult. The patient was continued on his anticoagulant medication on presentation. Many practices will hold anticoagulation for a period of days to weeks after stroke because of the risk of hemorrhagic conversion and utilize aspirin as secondary prevention during that time. Hemorrhagic conversion of the infarct was likely precipitated or worsened by the combination of antiplatelet and anticoagulant medications. Surgical decompression was lifesaving in this case. Note the fourth ventricular and aqueductal compression leading to obstructive hydrocephalus, which precedes brainstem compression.

between 6 and 12 weeks after the initial surgery. There is no evidence-based recommendation on the use of autologous bone flaps versus alloplastic implants, and the complex interactions between underlying disease, patient age, implant type and preservation method, and the timing of cranioplasty are incompletely understood.⁸² These measures, beyond typical postoperative wound care, are not required for patients who have suboccipital craniectomy because of the protection afforded by the fascial and muscular structures of the posterior neck. **FIGURE 3-6** provides a summary flowchart of the neurologic ICU management of patients after stroke from LVO who develop cerebral edema.

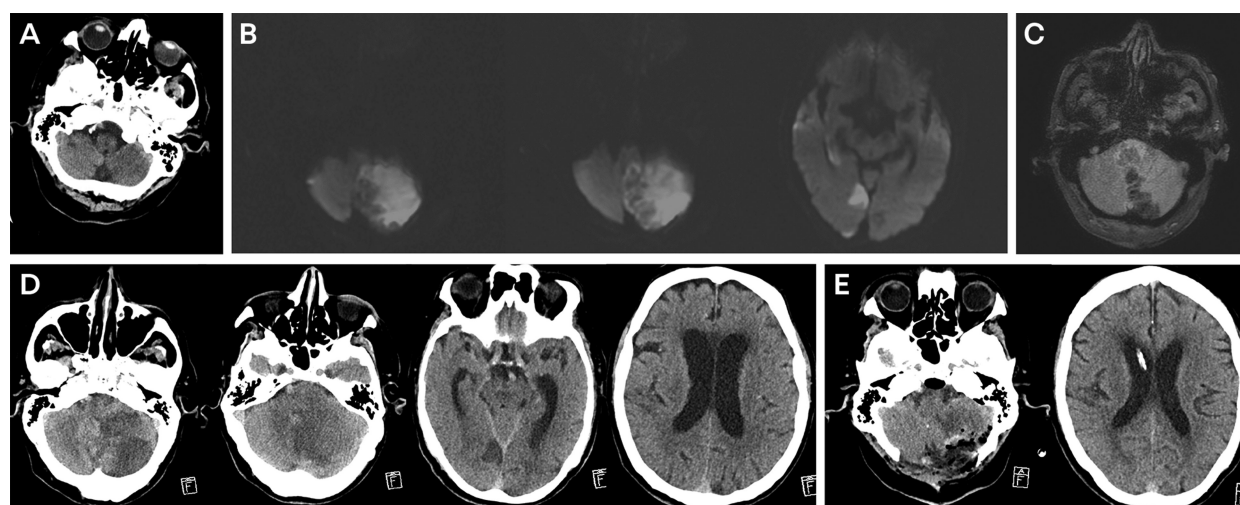


FIGURE 3-5 Cerebellar edema after stroke due to large vessel occlusion in the patient in **CASE 3-2**. **A**, Axial head CT shows subtle evidence of a left cerebellar hypodensity consistent with infarct. **B**, Axial diffusion-weighted MRI of the brain shows a large left cerebellar infarct with suspected hemorrhagic conversion and mass effect on the fourth ventricle and scattered bilateral occipital infarcts (right occipital infarct shown) consistent with embolic etiology. **C**, Axial gradient recalled echo (GRE) MRI of the brain confirms hemorrhagic transformation of the cerebellar infarct. **D**, Axial head CT 24 hours after initial presentation following stroke shows new effacement of the fourth ventricle and obstructive hydrocephalus. **E**, After urgent surgical decompression, axial head CT shows suboccipital craniectomy and left cerebellectomy with right external ventricular drain placement.

STROKE SECONDARY PREVENTION AFTER STROKE DUE TO LARGE VESSEL OCCLUSION

For patients with stroke due to LVO who undergo angiography, the neuroendovascular team in conjunction with the stroke team makes intraprocedural decisions about antiplatelet or anticoagulant use based on angiographic findings. The MR CLEAN-MED (safety and efficacy of aspirin, unfractionated heparin, both, or neither during endovascular stroke treatment) study,⁸⁶ which investigated the safety and efficacy of perioperative anticoagulation or antiplatelet therapy, suggests that routine perioperative anticoagulation or antiplatelet therapy is associated with an increased risk of symptomatic intracranial hemorrhage, without evidence for a beneficial effect on functional outcome. However, there are a range of exceptions, including patients who require cervical or intracranial stenting, in whom a more aggressive approach to antiplatelet therapy or anticoagulation might be taken, or patients who experience procedural complications, in whom such agents might not be given for multiple days.

After admission to the neurologic ICU and the performance of a protocolized evaluation to identify stroke etiology, the neurocritical care team makes decisions about secondary stroke prevention in conjunction with the stroke team and neuroendovascular team. This requires consideration of the patient's existing ischemic damage and the associated risk of hemorrhagic transformation as well as comorbidities. As a general rule, larger infarcts and deep supratentorial or brainstem infarcts are more prone to hemorrhage and demand greater caution when making decisions about the initiation of antiplatelet or anticoagulation agents.

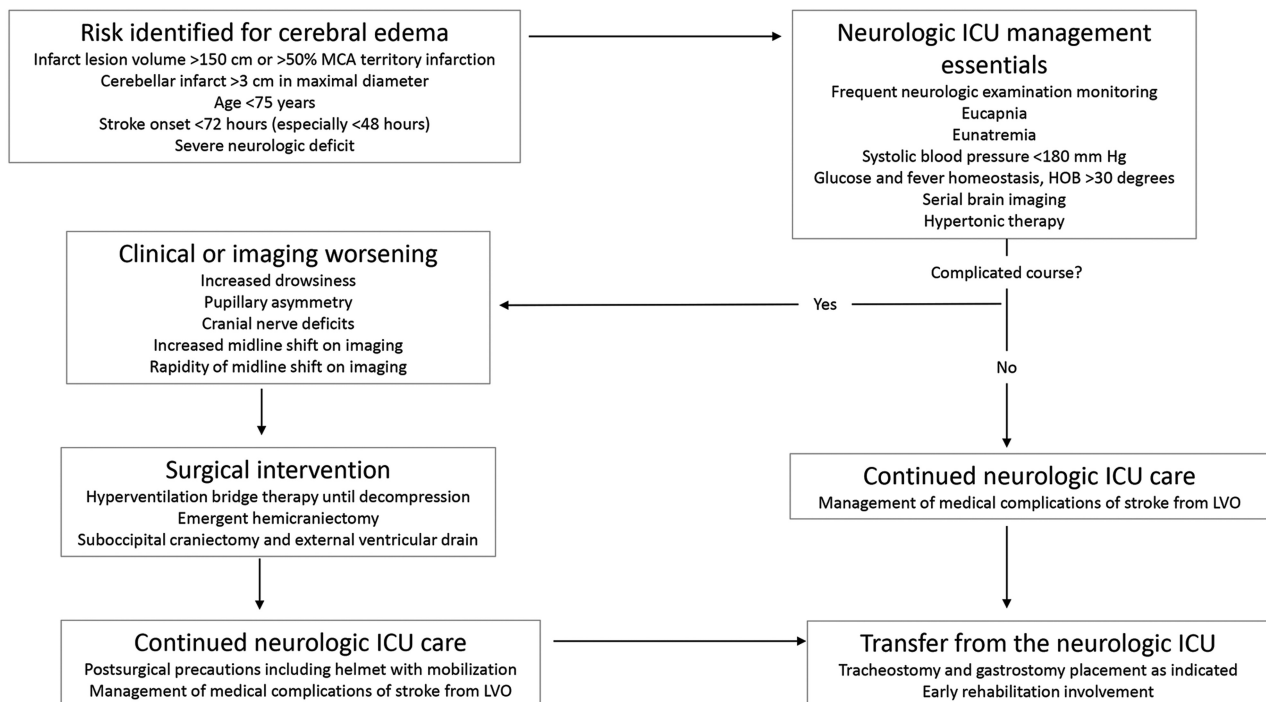


FIGURE 3-6

Flow diagram illustrating considerations in the management of cerebral edema in patients with large-volume cerebral infarction after stroke due to large vessel occlusion (LVO).

HOB = head of bed; ICU = intensive care unit; MCA = middle cerebral artery.

Antiplatelet Therapy

There are a range of antiplatelet options and doses for secondary stroke prophylaxis,⁸⁷ but aspirin remains the most studied and used. Aspirin should be started as soon as possible in all patients with acute stroke who do not have a contraindication, even if they were on aspirin before hospital admission. Potential reasons to stop aspirin use initially include thrombolytic treatment (aspirin use should be stopped for 24 hours), the potential need for surgery, such as craniectomy, or in the setting of hemorrhagic conversion.

Although aspirin plus clopidogrel is recommended for 21 to 30 days after a transient ischemic attack or minor stroke,^{88,89} this recommendation has not been expanded to include patients with stroke due to LVO. However, for patients with stroke from intracranial atherosclerosis, dual antiplatelet therapy is recommended and should be initiated as soon as possible after stroke and continued for 90 days in combination with other risk factor modifications as per the SAMMPRIS (Stenting and Aggressive Medical Management for Preventing Recurrent Stroke in Intracranial Stenosis) trial protocol.⁹⁰ Patients with carotid atherosclerosis may also be started on dual antiplatelet therapy for secondary stroke prevention while awaiting definitive management with a stent or endarterectomy,⁹¹ as might patients with ischemic stroke due to atrial fibrillation who are unable to initially take oral anticoagulation. The role of newer antiplatelet agents, and antiplatelet resistance and testing, remains a work in progress.⁸⁷

Anticoagulation

AHA/ASA guideline recommendations are to initiate anticoagulation for patients with a valid indication within 4 to 14 days of stroke onset.³⁵ When anticoagulation is indicated, direct oral anticoagulants are favored over warfarin based on multiple trials showing greater safety and easier administration,⁹² unless a patient has a particular indication for warfarin (eg, mechanical cardiac valve). Data from the 2023 ELAN (Early versus Late initiation of direct oral Anticoagulants in postischemic stroke patients with atrial fibrillation) trial compared early anticoagulation (eg, within 48 hours after a minor or moderate stroke or on day 6 or 7 after a major stroke) versus later anticoagulation (eg, day 3 or 4 after a minor stroke, day 6 or 7 after a moderate stroke, or day 12, 13, or 14 after a major stroke). Patients were stratified by imaging criteria, consistent with contemporary decision-making. Supporting guideline recommendations, early initiation of anticoagulation with direct oral anticoagulants was not associated with increased bleeding risk, but it did decrease the risk of recurrent ischemic stroke and systemic emboli.⁹³

In the neurologic ICU, patients initiated on anticoagulation are often commenced on heparin infusions as a bridge to oral anticoagulation, with the rationale that reversal is possible with protamine in the event of a hemorrhagic complication. These patients are typically imaged 24 hours after their doses of heparin have reached a therapeutic level, or sooner if there is a change in their neurologic examination, to assess for hemorrhage. Although reasonable, the utility of these measures as a universal approach is not clear.

SPECIAL POPULATIONS OF PATIENTS WITH STROKE DUE TO LARGE VESSEL OCCLUSION

There are additional populations of patients with stroke due to LVO for whom there are specific neurocritical care considerations.

KEY POINT

● After admission to the neurologic ICU and the performance of a protocolized evaluation to identify stroke etiology, decisions about secondary stroke prevention require consideration of the patient's existing ischemic damage and the associated risk of hemorrhagic transformation as well as comorbidities.

Large Infarct Core

Historically, patients with large infarct cores have not been offered thrombectomy, but they have been cared for in the neurologic ICU to manage poststroke edema and other complications. These patients have represented as much as 40% of the transfer population in some centers.⁹⁴ Randomized clinical trials have demonstrated the unequivocal benefit of thrombectomy for patients with large pretreatment-established core infarct volumes (as large as 150 mL).^{5,95,96} These patients experience improved independent functional outcomes and a halving of the risk of being bedbound with a modified Rankin Scale score of 5. Although rates of intracranial hemorrhage were higher, there was no evidence of a statistically significant increase in hemorrhagic complications. These data may lead to simplified thrombectomy selection criteria (ie, selection based solely on Alberta Stroke Program Early CT Score and evidence of an LVO) and subsequent higher rates of treatment for all patients with LVO.⁹⁷ All typical neurocritical care management considerations apply, although historical therapeutic nihilism regarding prognostication for these patients will need to evolve to reflect this contemporary therapeutic shift.

Low National Institutes of Health Stroke Scale Scores

The optimal treatment for patients with stroke due to LVO and low NIHSS scores is unclear. A significant proportion of these patients subsequently decompensate and either require rescue thrombectomy or experience significant disability as a result, with as many as 25% failing to achieve functional independence at 90 days.^{98,99} Randomized clinical trials are ongoing to determine the role of thrombectomy. Patients with stroke due to LVO and low NIHSS scores should be admitted to the neurologic ICU for close monitoring, regardless of whether thrombectomy is performed.

If thrombectomy is not performed, there should be close communication between the neurocritical care and stroke and neuroendovascular teams regarding management. Arterial blood pressure should be monitored continuously, and there should be ongoing evaluation for blood pressure–dependent examination changes for at least the first 24 hours. Crucially, a blood pressure floor needs to be set, typically determined by the lowest recorded blood pressure at which the patient is asymptomatic or, if some symptoms are unchanging, minimally symptomatic. Fluids should be administered to target euvolemia. Nocturnal hypotension due to increased vagal tone during sleep and orthostatic postural change represent the greatest risk of clinical deterioration. If not performed on initial presentation, delayed thrombectomy is an option for these patients if their collaterals do fail,¹⁰⁰ and it should be performed expediently for patients with gradual stepwise deterioration.

Seizure

Seizure may complicate any supratentorial stroke with cortical involvement. The risk is highest in patients with malignant cerebral edema, particularly after hemicraniectomy.¹⁰¹ Routine poststroke antiseizure medications and monitoring with EEG are not indicated, but clinicians should have a high suspicion for seizures in patients with a discordant or fluctuating neurologic examination without evidence of stroke progression, hemorrhagic conversion, or worsening edema, and consider EEG monitoring for these patients. Management is similar to that for seizures in other contexts.

MEDICAL COMPLICATIONS AFTER STROKE DUE TO LARGE VESSEL OCCLUSION

A range of medical complications occur after stroke due to LVO. Three specific examples are detailed in the following sections. Meticulous neurocritical care is necessary to limit the impact of these and other medical complications on patient outcomes.

Respiratory Failure

Given the degree of neurologic compromise and the potential need for thrombectomy or craniectomy, patients with stroke due to LVO are often intubated. Airway management decisions in this patient population may represent a clinical challenge. Classical weaning criteria are not reliably predictive of extubation success. When considering extubation, the most relevant factors are the patient's level of arousal and interaction, airway safety, and secretion handling. Early extubation may be pursued for patients with neurologic improvement, but reintubation rates of 16% to 37% have been reported.^{102,103}

Despite advances in noninvasive ventilatory support, these modalities are rarely applied to patients with stroke due to LVO because they often do not have adequate airway protection reflexes. Patients unable to protect their airway and maintain ventilation may require tracheostomy. Decisions about the timing for tracheostomy were evaluated in 2022 in the SETPOINT2 (Effect of Early versus Standard Approach to Tracheostomy on Functional Outcome at 6-months among Patients with Severe Stroke Receiving Mechanical Ventilation) trial. Among patients with severe stroke who required mechanical ventilation, a strategy of early tracheostomy (within 5 days of hospital admission), compared with a standard approach to tracheostomy (after 10 days of hospital admission), did not significantly improve the rate of survival without severe disability at 6 months.¹⁰⁴ However, tracheostomy does potentially confer benefits, such as reduced sedation needs, accelerated ventilator weaning, earlier mobilization and rehabilitation unit transfer, easier pulmonary hygiene, and fewer ICU complications.^{105,106}

Dysphagia

Adequate hydration and nutrition after stroke are crucial, but this may be restricted by oropharyngeal dysphagia, which affects 23% to 50% of patients with stroke due to LVO, heightening risks for aspiration, pneumonia, malnutrition, and worsening quality of life. In addition to oropharyngeal dysphagia, acute stroke is associated with dysmotility of the upper gastrointestinal tract, with delayed gastric emptying and gastroesophageal reflux. All patients with stroke should undergo an early nurse-led swallow assessment, with subsequent evaluation by speech therapy, with consideration of a fiber optic evaluation if there is concern for aspiration. Patients should have nothing by mouth until safe swallowing is confirmed. Enteral feeding should be prioritized early, with the FOOD (Feed Or Ordinary Diet in stroke) trial showing a 5.8% mortality benefit in patients with stroke if feeding was initiated within 7 days of admission.¹⁰⁷ The OPENS (OPTimizing Early enteral Nutrition in severe Stroke) trial found no significant difference in outcome at 90 days with the use of three different feeding strategies after severe stroke: (1) full enteral nutrition, (2) full enteral nutrition with prokinetic agents, or (3) hypocaloric enteral nutrition. Patients

KEY POINTS

- Routine poststroke antiseizure medications and monitoring with EEG in patients with stroke due to LVO is not indicated, but clinicians should have a high suspicion for seizures in patients with a discordant or fluctuating neurologic examination without evidence of stroke progression, hemorrhagic conversion or worsening edema, and EEG monitoring applied for these patients.
- When considering extubation of patients with stroke due to LVO, the most relevant factors are the patient's level of arousal and interaction, airway safety, and secretion handling.

who received full enteral nutrition with prokinetic drugs added had nonsignificant evidence of less gastric retention compared with patients in the full-enteral nutrition group. Rates of pneumonia were the same between groups. The trial terminated early because of excess mortality in the hypocaloric nutrition arm, reinforcing the importance of early feeding.¹⁰⁸ Questions therefore remain about the optimal approach to feeding, beyond meeting caloric needs early and completely. Initially, enteral nutrition should be delivered via a nasogastric tube in patients with swallowing difficulties, with percutaneous endoscopic gastrostomy offered for those with persistent dysphagia.

Deep Vein Thrombosis Prophylaxis

Deep vein thrombosis (DVT) is common after stroke. The CLOTS (Clots in Legs Or sTockings after Stroke) 3 trial demonstrated reduced DVT formation and improved 6-month survival rates (but not functional outcomes) with intermittent pneumatic compression devices.¹⁰⁹ Current AHA/ASA guidelines recommend intermittent pneumatic compression in immobile patients with stroke who do not have contraindications.³⁵ Although pharmacologic prophylaxis reduces the rate of DVT and pulmonary embolism in patients with stroke due to LVO, it carries the increased risk of intracranial and extracranial bleeding. In 2019, general ICU data suggest that adjunctive intermittent pneumatic compression does not result in a significantly lower incidence of proximal lower limb DVT compared with pharmacologic thromboprophylaxis alone.¹¹⁰ The relevance of these data to patients with stroke due to LVO is unclear. Identified risk factors for DVT in patients with stroke include prior venous thromboembolism, cancer, prestroke disability, lower extremity weakness, larger infarct volume, infection, longer length of hospital stay, biochemical indices of dehydration, and laboratory values with elevated levels of D-dimer, C-reactive protein, and homocysteine.¹¹¹ Timing, agent, and dose for pharmacologic prophylaxis remain variable in practice, but data from the PREVAIL (Efficacy and Safety of Enoxaparin Versus Unfractionated Heparin for the Prevention of Venous Thromboembolism After Acute Ischaemic Stroke) trial favor unfractionated heparin for patients with LVO, started 48 hours from stroke onset.¹¹²

REHABILITATION AFTER STROKE DUE TO LARGE VESSEL OCCLUSION

The ultimate goal of rehabilitation services for patients is to optimize independence in daily activities, improve quality of life, and facilitate a safe and effective transition back to their home and community. Acute rehabilitation should start early and be provided by organized, interprofessional teams in the neurologic ICU at a frequency and intensity commensurate with tolerance and anticipated benefit.³⁵ Rehabilitation services together offer assessment, goal setting, intervention or treatment planning and execution, support for inpatient rehabilitation placement, discharge planning, home modification recommendations, and emotional support and education services. Physical medicine and rehabilitation physicians increasingly play a central role in overseeing this range of care for patients in the neurologic ICU.¹¹³

For patients with stroke due to LVO who have depressed consciousness, a trial of a stimulant medication is reasonable to improve wakefulness and participation.¹¹⁴ A stimulant is typically initiated 5 to 7 days after the onset of the stroke and continued for at least 2 weeks. Dosing should be early morning and

midday (ie, when wakefulness is needed) and not twice a day with an evening dose (when sleep reinforcement is indicated). There are several stimulants,¹¹⁵ but after stroke, amantadine and modafinil are used most.

Patients ideally are prepared to be discharged from the hospital to a rehabilitation facility, as opposed to a skilled nursing facility. Despite similar conditions at discharge (ie, similar infarct volume and functional scores), patients discharged to skilled nursing facilities had worse outcomes compared with those discharged to inpatient rehabilitation facilities.¹¹⁶ Only 25% of patients triaged to skilled nursing facilities achieved a favorable outcome (modified Rankin Scale score of 0 to 2) compared with 46% of those who were exposed to inpatient rehabilitation ($P = .023$). Advocacy for patients with stroke due to LVO to have the most favorable disposition is crucial to ensure the best chance of recovery.

The role of early mobilization after stroke remains uncertain. Trials have failed to uniformly show consistent benefits in functional outcomes or quality of life.¹¹⁷ High-intensity early mobilization (ie, sitting, standing, walking) provided in at least three out-of-bed sessions within 24 hours of stroke onset compared with usual care reduced the odds of a favorable 3-month outcome,¹¹⁸ indicating that mobilization too early carries the risk of harm. Practice currently favors 24 hours of bedrest and hemodynamic stability after stroke before activity is attempted, with the goal of initiating activity within 3 days of admission and a focus on sitting, standing, and walking activity. Besides physical therapy, occupational therapy assists with splinting in patients with severe deficits, and functional exercises in those who are able. Speech and language therapy improves functional communication in patients with poststroke aphasia. Families who are at the bedside frequently can be educated about how they can assist with rehabilitation efforts. Together, the rehabilitation team provides tremendous benefits for patients with stroke due to LVO and their families.

Beyond the rehabilitation team, social work and case management services are essential in guiding discharge-planning coordination. These teams assist patients with stroke due to LVO and their families in navigating their new disability.

FAMILY COMMUNICATION AND NEUROPALLIATIVE CARE AFTER STROKE DUE TO LARGE VESSEL OCCLUSION

Because stroke due to LVO can cause significant disability, communication and prognostication are important to patients and their families. Compared with other areas of neurocritical care, there is a large body of literature to inform prognostication for patients with stroke due to LVO, including large core thrombectomy trials, but personalizing these data to the individual patient remains challenging. Therefore, prognostication after stroke due to LVO is highly variable across clinicians and institutions.

As a general rule, early poststroke nihilism should be avoided, and it is reasonable to allow several days for patients' courses and examinations to evolve. Messaging must be consistent across the entire care team. Family communication requires repeated interactions to build rapport and trust. Having family join daily rounds can be helpful.¹¹⁹

For patients with diminished capacity, shared decision-making in which surrogates and clinicians arrive together at an individualized decision based on patient values and preferences can improve clinician-family communication and ensure that patients receive the treatments they would choose.¹²⁰ Clear

KEY POINTS

- All patients with stroke should undergo an early nurse-led swallow assessment, subsequent evaluation by speech therapy, and consideration of a fiber optic evaluation if there is concern for aspiration. Patients should have nothing by mouth until safe swallowing is confirmed.
- Identified risk factors for deep venous thrombosis (DVT) in patients with stroke include prior venous thromboembolism, cancer, prestroke disability, lower extremity weakness, larger infarct volume, infection, longer length of hospital stay, biochemical indices of dehydration, and laboratory values with elevated levels of D-dimer, C-reactive protein, and homocysteine.
- Short-term rehabilitation for patients who have had stroke should start early and be provided by organized, interprofessional teams in the neurologic ICU, at a frequency and intensity commensurate with tolerance and anticipated benefit.

care-decision milestones should be inferred from each patient's life, including explicit or implicit positions on surgical decompression, tracheostomy and gastrostomy, living with nursing needs, and severe disability. Structured family communication with decision-aid support allows appropriate decision-making.¹²¹ There are various approaches to family communication, including authoritative (eg, recommending treatments without discussing values and preferences), informational (eg, disclosing just the prognosis without further discussions), advisory (eg, disclosing prognosis followed by discussion of values and preferences), and responsive (eg, eliciting values and preferences, then disclosing prognosis); the advisory approach is most common.¹²² For a detailed discussion of these approaches, refer to the article "Prognostication in Neurocritical Care" by Susanne Muehlschlegel, MD, MPH, FNCS, FCCM, FAAN,¹²³ in this issue of *Continuum*.

Palliative care services are used by neurointensivists in less than 11% of patients, according to a 2020 survey. Goals of care discussions and treatment decisions were the overwhelming reason for palliative care involvement.¹²⁴ A lack of referrals and delays in referral impede patient access to palliative care,¹²⁵ so inclusion of this team in the care of patients with stroke due to LVO should be considered as appropriate.

FUTURE STATE OF NEUROCRITICAL CARE FOR PATIENTS WITH STROKE DUE TO LARGE VESSEL OCCLUSION

Although predicting the future is fraught with uncertainty, trends help inform the future state of neurocritical care for patients with LVO. In the era of highly effective reperfusion therapy, neuroprotection has gained renewed energy. Clinical trials are actively evaluating potential neuroprotectant options.¹²⁶ The neurologic ICU will play a central role in the testing, validation, administration, and monitoring of patients for novel neuroprotectant agents.

Neurorehabilitation is gaining increased sophistication. Robotic and pharmacologic options hold promise in enabling patients to both start and expand rehabilitation in the intensive care environment.¹²⁷

The reach of telemedicine has expanded immensely following the COVID-19 pandemic. As an effective technique to disseminate localized expertise, "teleneuroICU" is likely to become a significant offering within the teleneurology landscape. The practice of virtually consulting on patients in an ICU setting with neurologic and neurosurgical conditions will be unrestricted by geography and, therefore, potentially offer service to patients with stroke due to LVO across the globe.¹²⁸ For more information about telemedicine, refer to the article "Teleneurology and Artificial Intelligence in Clinical Practice" by Elaine C. Jones, MD, FAAN, Benjamin R. Kummer, MD, and Jayne R. Wilkinson, MD, MSCE,¹²⁹ in this issue of *Continuum*.

Any predictions about the future include the anticipated role that artificial intelligence will play in augmenting the care provided by neurocritical care teams to patients with stroke due to LVO. In the neurologic ICU, the collection and analysis in real time of large datasets can play a crucial role in advancing the field and improving personalized patient care.¹³⁰

It is our increasing technologic and medical sophistication, combined with humanism, empathy, and compassion, that will continue to ensure that patients with stroke due to LVO are provided with the most appropriate and effective care and achieve the best possible outcomes.

CONCLUSION

For patients with stroke due to LVO, neurocritical care needs are complex and broad, from transfer for treatment consideration to rehabilitation initiation. The neurologic ICU plays a central role in the care of these individuals. A comprehensive, multidisciplinary, coordinated approach to care is necessary for the best patient outcomes.

REFERENCES

- 1 Leslie-Mazwi T, Chandra RV, Baxter BW, et al. ELVO: an operational definition. *J Neurointerv Surg* 2018;10(6):507-509. doi:10.1136/neurintsurg-2018-013792
- 2 Goyal M, Menon BK, van Zwam WH, et al. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. *Lancet* 2016;387(10029):1723-1731. doi:10.1016/S0140-6736(16)00163-X
- 3 Albers GW, Marks MP, Kemp S, et al. Thrombectomy for stroke at 6 to 16 hours with selection by perfusion imaging. *N Engl J Med* 2018;378(8):708-718. doi:10.1056/NEJMoA1713973
- 4 Nogueira RG, Jadhav AP, Haussen DC, et al. Thrombectomy 6 to 24 hours after stroke with a mismatch between deficit and infarct. *N Engl J Med* 2018;378(1):11-21. doi:10.1056/NEJMoA1706442
- 5 Yoshimura S, Nakagawa S, Takahashi T. FTY720 exacerbates blood-brain barrier dysfunction induced by IgG derived from patients with NMO and MOG disease. *Neurotox Res* 2021;39(4):1300-1309. doi:10.1007/s12640-021-00373-7
- 6 Menon BK, Buck BH, Singh N, et al. Intravenous tenecteplase compared with alteplase for acute ischaemic stroke in Canada (AcT): a pragmatic, multicentre, open-label, registry-linked, randomised, controlled, non-inferiority trial. *Lancet* 2022;400(10347):161-169. doi:10.1016/S0140-6736(22)01054-6
- 7 Silva GS, Rocha E, Mehta A, Sharrief A. Racial and ethnic diversity in endovascular thrombectomy trials. *Stroke Vasc Interv Neurol* 2024;4(1):e000613. doi:10.1161/SVIN.123.000613
- 8 Man S, Solomon N, Mac Grory B, et al. Trends in stroke thrombolysis care metrics and outcomes by race and ethnicity, 2003-2021. *JAMA Netw Open* 2024;7(2):e2352927. doi:10.1001/jamanetworkopen.2023.52927
- 9 Metcalf D, Zhang D. Racial and ethnic disparities in the usage and outcomes of ischemic stroke treatment in the United States. *J Stroke Cerebrovasc Dis* 2023;32(12):107393. doi:10.1016/j.jstrokecerebrovasdis.2023.107393
- 10 Fifi JT, Nguyen TN, Song S, et al. Sex differences in endovascular thrombectomy outcomes in large vessel occlusion: a propensity-matched analysis from the SELECT study. *J Neurointerv Surg* 2023;15(2):105-112. doi:10.1136/neurintsurg-2021-018348
- 11 Volpe SG, Zuniga MC, Caunca MR, Rosendale N. Gender and sex equity in stroke research, education, and care [published January 23, 2023]. *Stroke*. doi:10.1161/STROKEAHA.122.039893
- 12 Feigin VL, Owolabi MO, World Stroke Organization-Lancet Neurology Commission Stroke Collaboration Group. Pragmatic solutions to reduce the global burden of stroke: a World Stroke Organization-Lancet Neurology Commission. *Lancet Neurol* 2023;22(12):1160-1206. doi:10.1016/S1474-4422(23)00277-6
- 13 Wahlster S, Johnson NJ. The neurocritical care examination and workup. *Continuum (Minneapolis)* 2024;30(3, Neurocritical Care):556-587.
- 14 Dusenbury W, Mathiesen C, Whaley M, et al. Ideal foundational requirements for stroke program development and growth: a scientific statement from the American Heart Association. *Stroke* 2023;54(4):e175-e187. doi:10.1161/STR.0000000000000424
- 15 Moheet AM, Livesay SL, Abdelhak T, et al. Standards for neurologic critical care units: a statement for healthcare professionals from the Neurocritical Care Society. *Neurocrit Care* 2018;29(2):145-160. doi:10.1007/s12028-018-0601-1
- 16 Rezek MA, Murray E, Youngren MN, Durham NT, Michael SS. Door-to-imaging time for acute stroke patients is adversely affected by emergency department crowding. *Stroke* 2017;48(1):49-54. doi:10.1161/STROKEAHA.116.015131
- 17 Regenhardt RW, Rosenthal JA, Dmytriw AA, et al. Direct to angio-suite large vessel occlusion transfers achieve faster arrival-to-puncture times and improved outcomes. *Stroke Vasc Interv Neurol* 2022;2(6):e000327. doi:10.1161/svin.121.000327
- 18 Kuc A, Isenberg DL, Kraus CK, et al. Factors associated with door-in-door-out times in large vessel occlusion stroke patients undergoing endovascular therapy. *Am J Emerg Med* 2023;69:87-91. doi:10.1016/j.ajem.2023.04.009

- 19 Regenhardt RW, Awad A, Kraft AW, et al. Characterizing reasons for stroke thrombectomy ineligibility among potential candidates transferred in a hub-and-spoke network. *Stroke Vasc Interv Neurol* 2022;2(5):e000282. doi:10.1161/svin.121.000282
- 20 Colvin MO, Eisen LA, Gong MN. Improving the patient handoff process in the intensive care unit: keys to reducing errors and improving outcomes. *Semin Respir Crit Care Med* 2016;37(1):96-106. doi:10.1055/s-0035-1570351
- 21 Wu Y, Wu X, Pan H, et al. Does prolonged emergency department length of stay (EDLOS) affect the outcomes of acute ischemic stroke patients? *Am J Emerg Med* 2021;45:258-263. doi:10.1016/j.ajem.2020.08.061
- 22 Rincon F, Mayer SA, Rivolta J, et al. Impact of delayed transfer of critically ill stroke patients from the emergency department to the neuro-ICU. *Neurocrit Care* 2010;13(1):75-81. doi:10.1007/s12028-010-9347-0
- 23 Murray NM, Joshi AN, Kronfeld K, et al. A standardized checklist improves the transfer of stroke patients from the neurocritical care unit to hospital ward. *Neurohospitalist* 2020;10(2):100-108. doi:10.1177/1941874419873810
- 24 Forti RM, Favilla CG, Cochran JM, et al. Transcranial optical monitoring of cerebral hemodynamics in acute stroke patients during mechanical thrombectomy. *J Stroke Cerebrovasc Dis* 2019;28(6):1483-1494. doi:10.1016/j.jstrokecerebrovasdis.2019.03.019
- 25 Shah K, Clark A, Desai SM, Jadhav AP. Causes, predictors, and timing of early neurological deterioration and symptomatic intracranial hemorrhage after administration of IV tPA. *Neurocrit Care* 2022;36(1):123-129. doi:10.1007/s12028-021-01266-5
- 26 Leonardi-Bee J, Bath PMW, Phillips SJ, Sandercock PAG, IST Collaborative Group. Blood pressure and clinical outcomes in the International Stroke Trial. *Stroke* 2002;33(5):1315-1320. doi:10.1161/01.str.0000014509.11540.66
- 27 Malhotra K, Goyal N, Katsanos AH, et al. Association of blood pressure with outcomes in acute stroke thrombectomy. *Hypertension* 2020;75(3):730-739. doi:10.1161/HYPERTENSIONAHA.119.14230
- 28 John S, Hazaa W, Uchino K, Hussain MS. Timeline of blood pressure changes after intra-arterial therapy for acute ischemic stroke based on recanalization status. *J Neurointerv Surg* 2017;9(5):455-458. doi:10.1136/neurintsurg-2016-012369
- 29 Mazighi M, Richard S, Lapergue B, et al. Safety and efficacy of intensive blood pressure lowering after successful endovascular therapy in acute ischaemic stroke (BP-TARGET): a multicentre, open-label, randomised controlled trial. *Lancet Neurol* 2021;20(4):265-274. doi:10.1016/S1474-4422(20)30483-X
- 30 Manning LS, Rothwell PM, Potter JF, Robinson TG. Prognostic significance of short-term blood pressure variability in acute stroke: systematic review. *Stroke* 2015;46(9):2482-2490. doi:10.1161/STROKEAHA.115.010075
- 31 Jeong HG, Kim BJ, Kim H, et al. Blood pressure drop and penumbral tissue loss in nonrecanalized emergent large vessel occlusion. *Stroke* 2019;50(10):2677-2684. doi:10.1161/STROKEAHA.119.025426
- 32 Singh RJ, Doshi D, Barber PA. Hypoglycemia causing focal cerebral hypoperfusion and acute stroke symptoms. *Can J Neurol Sci* 2021;48(4):550-552. doi:10.1017/cjn.2020.246
- 33 Kruyt ND, Biessels GJ, Devries JH, Roos YB. Hyperglycemia in acute ischemic stroke: pathophysiology and clinical management. *Nat Rev Neurol* 2010;6(3):145-155. doi:10.1038/nrneurol.2009.231
- 34 Johnston KC, Bruno A, Pauls Q, et al. Intensive vs standard treatment of hyperglycemia and functional outcome in patients with acute ischemic stroke: the SHINE Randomized Clinical Trial. *JAMA* 2019;322(4):326-335. doi:10.1001/jama.2019.9346
- 35 Powers WJ, Rabinstein AA, Ackerson T, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2019;50(12):e344-e418. doi:10.1161/STR.0000000000000211
- 36 Phipps MS, Desai RA, Wira C, Bravata DM. Epidemiology and outcomes of fever burden among patients with acute ischemic stroke. *Stroke* 2011;42(12):3357-3362. doi:10.1161/STROKEAHA.111.621425
- 37 Horn CM, Sun CHJ, Nogueira RG, et al. Endovascular reperfusion and cooling in cerebral acute ischemia (reCLAIM I). *J Neurointerv Surg* 2014;6(2):91-95. doi:10.1136/neurintsurg-2013-010656
- 38 Leslie-Mazwi T, Chen M, Yi J, et al. Post-thrombectomy management of the ELVO patient: guidelines from the Society of NeuroInterventional Surgery. *J Neurointerv Surg* 2017;9(12):1258-1266. doi:10.1136/neurintsurg-2017-013270
- 39 Krishnan R, Mays W, Eljovich L. Complications of mechanical thrombectomy in acute ischemic stroke. *Neurology* 2021;97(20 Suppl 2):S115-S125. doi:10.1212/WNL.00000000000012803
- 40 Pilgram-Pastor SM, Piechowiak EI, Dobrocky T, et al. Stroke thrombectomy complication management. *J Neurointerv Surg* 2021;13(10):912-917. doi:10.1136/neurintsurg-2021-017349

- 41 Happi Ngankou E, Gory B, Marnat G, et al. Thrombectomy complications in large vessel occlusions: incidence, predictors, and clinical impact in the ETIS registry. *Stroke* 2021;52(12): e764-e768. doi:10.1161/STROKEAHA.121.034865
- 42 Hendrix P, Collins MK, Goren O, et al. Femoral access-site complications with tenecteplase versus alteplase before mechanical thrombectomy for large-vessel-occlusion stroke. *AJNR Am J Neuroradiol* 2023;44(6): 681-686. doi:10.3174/ajnr.A7862
- 43 Khanna O, Velagapudi L, Das S, et al. A comparison of radial versus femoral artery access for acute stroke interventions. *Journal of Neurosurgery* 2020;135:1-6. doi:10.3171/2020.7.JNS201174
- 44 Jolly SS, Yusuf S, Cairns J, et al. Radial versus femoral access for coronary angiography and intervention in patients with acute coronary syndromes (RIVAL): a randomised, parallel group, multicentre trial. *Lancet* 2011;377(9775):1409-1420. doi:10.1016/S0140-6736(11)60404-2
- 45 Colombo E, Rinaldo L, Lanzino G. Direct carotid puncture in acute ischaemic stroke intervention. *Stroke Vasc Neurol* 2020;5(1):71-79. doi:10.1136/svn-2019-000260
- 46 Shapiro SZ, Sabacinski KA, Mantripragada K, et al. Access-site complications in mechanical thrombectomy for acute ischemic stroke: a review of prospective trials. *AJNR Am J Neuroradiol* 2020;41(3):477-481. doi:10.3174/ajnr.A6423
- 47 Catapano JS, Fredrickson VL, Fujii T, et al. Complications of femoral versus radial access in neuroendovascular procedures with propensity adjustment. *J Neurointerv Surg* 2020;12(6):611-615. doi:10.1136/neurintsurg-2019-015569
- 48 Wareham J, Luppe S, Youssef A, Crossley R, Mortimer A. Safety profile of an 8F femoral arteriotomy closure using the Angio-Seal device in thrombolysed acute stroke patients undergoing thrombectomy. *Interv Neuroradiol* 2018;24(5):540-545. doi:10.1177/1591019918779022
- 49 Beraldo de Andrade P, de Ribamar Costa J, Rinaldi FS, et al. Vascular closure devices attenuate femoral access complications of primary percutaneous coronary intervention. *J Invasive Cardiol* 2020;32(10):364-370.
- 50 Joshi KC, Khanna R, Beer-Furlan A, et al. Femoral access site complications following neurointerventional procedures: economic implications at a single center. *J Neurosurg* 2023; 139(4):1078-1082. doi:10.3171/2023.1.JNS222171
- 51 Terruso V, D'Amelio M, Di Benedetto N, et al. Frequency and determinants for hemorrhagic transformation of cerebral infarction. *Neuroepidemiology* 2009;33(3):261-265. doi:10.1159/000229781
- 52 Yang Y, Rosenberg GA. Blood-brain barrier breakdown in acute and chronic cerebrovascular disease. *Stroke* 2011;42(11):3323-3328. doi:10.1161/STROKEAHA.110.608257
- 53 Castellanos M, Leira R, Serena J, et al. Plasma metalloproteinase-9 concentration predicts hemorrhagic transformation in acute ischemic stroke. *Stroke* 2003;34(1):40-46.
- 54 Wang X, Tsuji K, Lee SR, et al. Mechanisms of hemorrhagic transformation after tissue plasminogen activator reperfusion therapy for ischemic stroke. *Stroke* 2004;35(11 Suppl 1): 2726-2730. doi:10.1161/01.STR.0000143219.16695.af
- 55 von Kummer R, Broderick JP, Campbell BCV, et al. The Heidelberg Bleeding Classification: classification of bleeding events after ischemic stroke and reperfusion therapy. *Stroke* 2015; 46(10):2981-2986. doi:10.1161/STROKEAHA.115.010049
- 56 Neuberger U, Möhlenbruch MA, Herweh C, et al. Classification of bleeding events: comparison of ECASS III (European Cooperative Acute Stroke Study) and the new Heidelberg Bleeding Classification. *Stroke* 2017;48(7):1983-1985. doi:10.1161/STROKEAHA.117.016735
- 57 Fiorelli M, Bastianello S, von Kummer R, et al. Hemorrhagic transformation within 36 hours of a cerebral infarct: relationships with early clinical deterioration and 3-month outcome in the European Cooperative Acute Stroke Study I (ECASS I) cohort. *Stroke* 1999;30(11):2280-2284. doi:10.1161/01.str.30.11.2280
- 58 Strbian D, Sairanen T, Meretoja A, et al. Patient outcomes from symptomatic intracerebral hemorrhage after stroke thrombolysis. *Neurology* 2011;77(4):341-348. doi:10.1212/WNL.0b013e3182267b8c
- 59 Lummel N, Schulte-Altedorneburg G, Bernau C, et al. Hyperattenuated intracerebral lesions after mechanical recanalization in acute stroke. *AJNR Am J Neuroradiol* 2014;35(2):345-351. doi:10.3174/ajnr.A3656
- 60 Almqvist H, Holmin S, Mazya MV. Dual energy CT after stroke thrombectomy alters assessment of hemorrhagic complications. *Neurology* 2019; 93(11):e1068-e1075. doi:10.1212/WNL.0000000000008093
- 61 Renú A, Amaro S, Laredo C, et al. Relevance of blood-brain barrier disruption after endovascular treatment of ischemic stroke: dual-energy computed tomographic study. *Stroke* 2015;46(3): 673-679. doi:10.1161/STROKEAHA.114.008147
- 62 Djurdjevic T, Rehwald R, Knoflach M, et al. Prediction of infarction development after endovascular stroke therapy with dual-energy computed tomography. *Eur Radiol* 2017;27(3): 907-917. doi:10.1007/s00330-016-4412-5
- 63 Bonatti M, Lombardo F, Zamboni GA, et al. Iodine extravasation quantification on dual-energy ct of the brain performed after mechanical thrombectomy for acute ischemic stroke can predict hemorrhagic complications. *AJNR Am J Neuroradiol* 2018;39(3):441-447. doi:10.3174/ajnr.A5513

- 64 Maier B, Desilles JP, Mazighi M. Intracranial hemorrhage after reperfusion therapies in acute ischemic stroke patients. *Front Neurol* 2020;11:599908. doi:10.3389/fneur.2020.599908
- 65 Battey TWK, Karki M, Singhal AB, et al. Brain edema predicts outcome after nonlacunar ischemic stroke. *Stroke* 2014;45(12):3643-3648. doi:10.1161/STROKEAHA.114.006884
- 66 Wijidicks EFM, Sheth KN, Carter BS, et al. Recommendations for the management of cerebral and cerebellar infarction with swelling: a statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke* 2014;45(4):1222-1238. doi:10.1161/01.str.0000441965.15164.d6
- 67 Krieger DW, Demchuk AM, Kasner SE, Jauss M, Hantson L. Early clinical and radiological predictors of fatal brain swelling in ischemic stroke. *Stroke* 1999;30(2):287-292. doi:10.1161/01.str.30.2.287
- 68 Cucchiara BL, Kasner SE, Wolk DA, et al. Early impairment in consciousness predicts mortality after hemispheric ischemic stroke. *Crit Care Med* 2004;32(1):241-245. doi:10.1097/01.CCM.0000104113.36384.81
- 69 Kasner SE, Demchuk AM, Berrouschot J, et al. Predictors of fatal brain edema in massive hemispheric ischemic stroke. *Stroke* 2001;32(9):2117-2123. doi:10.1161/hs0901.095719
- 70 Thomalla G, Hartmann F, Juettler E, et al. Prediction of malignant middle cerebral artery infarction by magnetic resonance imaging within 6 hours of symptom onset: a prospective multicenter observational study. *Ann Neurol* 2010;68(4):435-445. doi:10.1002/ana.22125
- 71 Kruetzelmann A, Hartmann F, Beck C, et al. Combining magnetic resonance imaging within six-hours of symptom onset with clinical follow-up at 24 h improves prediction of "malignant" middle cerebral artery infarction. *Int J Stroke* 2014;9(2):210-214. doi:10.1111/ijls.12060
- 72 Ng FC, Yassi N, Sharma G, et al. Cerebral edema in patients with large hemispheric infarct undergoing reperfusion treatment: a HERMES meta-analysis. *Stroke* 2021;52(11):3450-3458. doi:10.1161/STROKEAHA.120.033246
- 73 Jha RM, Raikwar SP, Mihaljevic S, et al. Emerging therapeutic targets for cerebral edema. *Expert Opin Ther Targets* 2021;25(11):917-938. doi:10.1080/14728222.2021.2010045
- 74 Shi J, Tan L, Ye J, Hu L. Hypertonic saline and mannitol in patients with traumatic brain injury. *Medicine (Baltimore)* 2020;99(35):e21655. doi:10.1097/MD.00000000000021655
- 75 Podell JE, Morris NA. Traumatic brain injury and traumatic spinal cord injury. *Continuum (Minneapolis)* 2024;30(3, Neurocritical Care):721-756.
- 76 Su Y, Liu Y, Chen Z, Cui L. Comparison of equiosmolar doses of 10% hypertonic saline and 20% mannitol for controlling intracranial hypertension in patients with large hemispheric infarction. *Clin Neurol Neurosurg* 2021;200:106359. doi:10.1016/j.clineuro.2020.106359
- 77 Christensen MS, Paulson OB, Olesen J, et al. Cerebral apoplexy (stroke) treated with or without prolonged artificial hyperventilation. 1. cerebral circulation, clinical course, and cause of death. *Stroke* 1973;4(4):568-631. doi:10.1161/01.str.4.4.568
- 78 Jüttler E, Schwab S, Schmiedek P, et al. Decompressive surgery for the treatment of malignant infarction of the middle cerebral artery (DESTINY): a randomized, controlled trial. *Stroke* 2007;38(9):2518-2525. doi:10.1161/STROKEAHA.107.485649
- 79 Hofmeijer J, Kappelle LJ, Algra A, et al. Surgical decompression for space-occupying cerebral infarction (the Hemicraniectomy After Middle Cerebral Artery infarction with Life-threatening Edema Trial [HAMLET]): a multicentre, open, randomised trial. *Lancet Neurol* 2009;8(4):326-333. doi:10.1016/S1474-4422(09)70047-X
- 80 Vahedi K, Hofmeijer J, Juettler E, et al. Early decompressive surgery in malignant infarction of the middle cerebral artery: a pooled analysis of three randomised controlled trials. *Lancet Neurol* 2007;6(3):215-222. doi:10.1016/S1474-4422(07)70036-4
- 81 Jüttler E, Unterberg A, Woitzik J, et al. Hemicraniectomy in older patients with extensive middle-cerebral-artery stroke. *N Engl J Med* 2014;370(12):1091-1100. doi:10.1056/NEJMoa1311367
- 82 Beez T, Munoz-Bendix C, Steiger HJ, Beseoglu K. Decompressive craniectomy for acute ischemic stroke. *Crit Care* 2019;23(1):209. doi:10.1186/s13054-019-2490-x
- 83 Jauss M, Krieger D, Hornig C, Schramm J, Busse O. Surgical and medical management of patients with massive cerebellar infarctions: results of the German-Austrian Cerebellar Infarction Study. *J Neurol* 1999;246(4):257-264. doi:10.1007/s004150050344
- 84 Kim MJ, Park SK, Song J, et al. Preventive suboccipital decompressive craniectomy for cerebellar infarction: a retrospective-matched case-control study. *Stroke* 2016;47(10):2565-2573. doi:10.1161/STROKEAHA.116.014078
- 85 St Louis EK, Wijidicks EF, Li H. Predicting neurologic deterioration in patients with cerebellar hematomas. *Neurology* 1998;51(5):1364-1369. doi:10.1212/wnl.51.5.1364
- 86 van der Steen W, van de Graaf RA, Chalos V, et al. Safety and efficacy of aspirin, unfractionated heparin, both, or neither during endovascular stroke treatment (MR CLEAN-MED): an open-label, multicentre, randomised controlled trial. *Lancet* 2022;399(10329):1059-1069. doi:10.1016/S0140-6736(22)00014-9

- 87 Kamarova M, Baig S, Patel H, et al. Antiplatelet use in ischemic stroke. *Ann Pharmacother* 2022; 56(10):1159-1173. doi:10.1177/10600280211073009
- 88 Dong J, Wang F, Sundararajan S. Use of dual antiplatelet therapy following ischemic stroke. *Stroke* 2020;51(5):e78-e80. doi:10.1161/STROKEAHA.119.028400
- 89 Bhatia K, Jain V, Aggarwal D, et al. Dual antiplatelet therapy versus aspirin in patients with stroke or transient ischemic attack: meta-analysis of randomized controlled trials. *Stroke* 2021;52(6):e217-e223. doi:10.1161/STROKEAHA.120.033033
- 90 Derdeyn CP, Chimowitz MI, Lynn MJ, et al. Aggressive medical treatment with or without stenting in high-risk patients with intracranial artery stenosis (SAMMPRIS): the final results of a randomised trial. *Lancet* 2014;383(9914):333-341. doi:10.1016/S0140-6736(13)62038-3
- 91 Wong KSL, Chen C, Fu J, et al. Clopidogrel plus aspirin versus aspirin alone for reducing embolisation in patients with acute symptomatic cerebral or carotid artery stenosis (CLAIR study): a randomised, open-label, blinded-endpoint trial. *Lancet Neurol* 2010;9(5):489-497. doi:10.1016/S1474-4422(10)70060-0
- 92 Carnicelli AP, Hong H, Connolly SJ, et al. Direct oral anticoagulants versus warfarin in patients with atrial fibrillation: patient-level network meta-analyses of randomized clinical trials with interaction testing by age and sex. *Circulation* 2022;145(4):242-255. doi:10.1161/CIRCULATIONAHA.121.056355
- 93 Fischer U, Koga M, Strbian D, et al. Early versus later anticoagulation for stroke with atrial fibrillation. *N Engl J Med* 2023;388(26):2411-2421. doi:10.1056/NEJMoa2303048
- 94 Leslie-Mazwi TM, Hirsch JA, Falcone GJ, et al. Endovascular stroke treatment outcomes after patient selection based on magnetic resonance imaging and clinical criteria. *JAMA Neurol* 2016; 73(1):43-49. doi:10.1001/jamaneurol.2015.3000
- 95 Sarraj A, Hassan AE, Abraham MG, et al. Trial of endovascular thrombectomy for large ischemic strokes. *N Engl J Med* 2023;388(14):1259-1271. doi:10.1056/NEJMoa2214403
- 96 Huo X, Ma G, Tong X, et al. Trial of endovascular therapy for acute ischemic stroke with large infarct. *N Engl J Med* 2023;388(14):1272-1283. doi:10.1056/NEJMoa2213379
- 97 Chen M, Leslie-Mazwi TM, Hirsch JA, Albuquerque FC. Large core stroke thrombectomy: paradigm shift or futile exercise? *J Neurointerv Surg* 2023;15(5):413-414. doi:10.1136/jnis-2023-020219
- 98 Haussen DC, Bousslama M, Grossberg JA, et al. Too good to intervene? Thrombectomy for large vessel occlusion strokes with minimal symptoms: an intention-to-treat analysis. *J Neurointerv Surg* 2017;9(10):917-921. doi:10.1136/neurintsurg-2016-012633
- 99 Lin CH, Saver JL, Ovbiagele B, et al. Effects of endovascular therapy for mild stroke due to proximal or M2 occlusions: meta-analysis. *J Neurointerv Surg* 2023;15(4):350-354. doi:10.1136/neurintsurg-2022-018662
- 100 Sarraj A, Kleinig TJ, Hassan AE, et al. Association of endovascular thrombectomy vs medical management with functional and safety outcomes in patients treated beyond 24 hours of last known well: the SELECT Late Study. *JAMA Neurol* 2023;80(2):172-182. doi:10.1001/jamaneurol.2022.4714
- 101 Creutzfeldt CJ, Tirschwell DL, Kim LJ, et al. Seizures after decompressive hemicraniectomy for ischaemic stroke. *J Neurol Neurosurg Psychiatry* 2014;85(7):721-725. doi:10.1136/jnnp-2013-305678
- 102 Steidl C, Bösel J, Suntrup-Krueger S, et al. Tracheostomy, extubation, reintubation: airway management decisions in intubated stroke patients. *Cerebrovasc Dis* 2017;44(1-2):1-9. doi:10.1159/000471892
- 103 Anderson CD, Bartscher JF, Scripko PD, et al. Neurologic examination and extubation outcome in the neurocritical care unit. *Neurocrit Care* 2011; 15(3):490-497. doi:10.1007/s12028-010-9369-7
- 104 Bösel J, Schönenberger S, Seder DB. Early vs standard approach to tracheostomy and functional outcome among patients with severe stroke receiving mechanical ventilation—reply. *JAMA* 2022;328(10):987-988. doi:10.1001/jama.2022.12667
- 105 Siempos II, Ntaidou TK, Filippidis FT, Choi AMK. Effect of early versus late or no tracheostomy on mortality and pneumonia of critically ill patients receiving mechanical ventilation: a systematic review and meta-analysis. *Lancet Respir Med* 2015;3(2):150-158. doi:10.1016/S2213-2600(15)00007-7
- 106 McCredie VA, Alali AS, Scales DC, et al. Effect of early versus late tracheostomy or prolonged intubation in critically ill patients with acute brain injury: a systematic review and meta-analysis. *Neurocrit Care* 2017;26(1):14-25. doi:10.1007/s12028-016-0297-z
- 107 Dennis M, Lewis S, Cranswick G, Forbes J, FOOD Trial Collaboration. FOOD: a multicentre randomised trial evaluating feeding policies in patients admitted to hospital with a recent stroke. *Health Technol Assess* 2006;10(2):iii-iv, ix-x, 1-120. doi:10.3310/hta10020
- 108 Zhao J, Yuan F, Song C, et al. Safety and efficacy of three enteral feeding strategies in patients with severe stroke in China (OPENS): a multicentre, prospective, randomised, open-label, blinded-endpoint trial. *Lancet Neurol* 2022;21(4):319-328. doi:10.1016/S1474-4422(22)00010-2

- 109 CLOTS (Clots in Legs Or sTockings after Stroke) Trials Collaboration, Dennis M, Sandercock P, et al. Effectiveness of intermittent pneumatic compression in reduction of risk of deep vein thrombosis in patients who have had a stroke (CLOTS 3): a multicentre randomised controlled trial. *Lancet* 2013;382(9891):516-524. doi:10.1016/S0140-6736(13)61050-8
- 110 Arabi YM, Al-Hameed F, Burns KEA, et al. Adjunctive intermittent pneumatic compression for venous thromboprophylaxis. *N Engl J Med* 2019;380(14):1305-1315. doi:10.1056/NEJMoal816150
- 111 Tøndel BG, Morelli VM, Hansen JB, Braekkan SK. Risk factors and predictors for venous thromboembolism in people with ischemic stroke: a systematic review. *J Thromb Haemost* 2022;20(10):2173-2186. doi:10.1111/jth.15813
- 112 Sherman DG, Albers GW, Bladin C, et al. The efficacy and safety of enoxaparin versus unfractionated heparin for the prevention of venous thromboembolism after acute ischaemic stroke (PREVAIL Study): an open-label randomised comparison. *Lancet* 2007;369(9570):1347-1355. doi:10.1016/S0140-6736(07)60633-3
- 113 Belagaje SR. Stroke rehabilitation. *Continuum (Minneapolis)* 2017;23(1, Cerebrovascular Disease):238-253. doi:10.1212/CON.0000000000000423
- 114 Leclerc AM, Riker RR, Brown CS, et al. Amantadine and modafinil as neurostimulants following acute stroke: a retrospective study of intensive care unit patients. *Neurocrit Care* 2021;34(1):102-111. doi:10.1007/s12028-020-00986-4
- 115 Barra ME, Edlow BL, Brophy GM. Pharmacologic therapies to promote recovery of consciousness. *Semin Neurol* 2022;42(3):335-347. doi:10.1055/s-0042-1755271
- 116 Belagaje SR, Sun CHJ, Nogueira RG, et al. Discharge disposition to skilled nursing facility after endovascular reperfusion therapy predicts a poor prognosis. *J Neurointerv Surg* 2015;7(2):99-103. doi:10.1136/neurintsurg-2013-011045
- 117 Cumming TB, Churilov L, Collier J, et al. Early mobilization and quality of life after stroke: Findings from AVERT. *Neurology* 2019;93(7):e717-e728. doi:10.1212/WNL.0000000000007937
- 118 Bernhardt J, Langhorne P, Lindley R. Efficacy and safety of very early mobilisation within 24 h of stroke onset (AVERT): a randomised controlled trial. *The Lancet* 2015;386(9988):46-55. doi:10.1016/S0140-6736(15)60690-0
- 119 Simon K, Sankara IR, Gioe C, Newcomb P. Including family members in rounds to improve communication in intensive care. *J Nurs Care Qual* 2021;36(1):25-31. doi:10.1097/NCQ.0000000000000483
- 120 Goostrey K, Muehlschlegel S. Prognostication and shared decision making in neurocritical care. *BMJ* 2022;377:e060154. doi:10.1136/bmj-2021-060154
- 121 Muehlschlegel S, Goostrey K, Flahive J, et al. A pilot randomized clinical trial of a goals-of-care decision aid for surrogates of severe acute brain injury patients. *Neurology* 2022;99(14):e1446-1455. doi:10.1212/WNL.000000000000200937
- 122 Ge C, Goss AL, Crawford S, et al. Variability of prognostic communication in critically ill neurologic patients: a pilot multicenter mixed-methods study. *Crit Care Explor* 2022;4(2):e0640. doi:10.1097/CCE.0000000000000640
- 123 Muehlschlegel S. Prognostication in neurocritical care. *Continuum (Minneapolis)* 2024;30(3, Neurocritical Care):878-903.
- 124 Bar B, Creutzfeldt CJ, Rubin MA. Palliative care in the neuro-ICU: perceptions, practice patterns, and preferences of neurointensivists. *Neurocrit Care* 2020;32(1):302-305. doi:10.1007/s12028-019-00838-w
- 125 Morris M, Mroz EL, Popescu C, Baron-Lee J, Busl KM. Palliative care services in the neuroICU: opportunities and persisting barriers. *Am J Hosp Palliat Care* 2021;38(11):1342-1347. doi:10.1177/1049909120987215
- 126 NoNO Inc. A multicentre, randomized, double-blinded, placebo-controlled, parallel group, single-dose design to determine the efficacy and safety of nerinetide in participants with acute ischemic stroke undergoing endovascular thrombectomy excluding thrombolysis. [clinicaltrials.gov; 2023](https://clinicaltrials.gov/study/NCT04462536). Accessed March 27, 2024. clinicaltrials.gov/study/NCT04462536
- 127 Fazekas G, Tavaszi I. The future role of robots in neuro-rehabilitation. *Expert Rev Neurother* 2019;19(6):471-473. doi:10.1080/14737175.2019.1617700
- 128 Freeman WD, Rogers A, Rabinstein A. TeleNeuroICU: expanding the reach of subspecialty neurocritical care. *Semin Neurol* 2022;42(1):18-30. doi:10.1055/s-0041-1742093
- 129 Jones EC, Kummer BR, Wilkinson JR. Teleneurology and artificial intelligence in clinical practice. *Continuum (Minneapolis)* 2024;30(3, Neurocritical Care):904-914.
- 130 Beam AL, Drazen JM, Kohane IS, et al. Artificial intelligence in medicine. *N Engl J Med* 2023;388(13):1220-1221. doi:10.1056/NEJMe2206291